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(54) **METHOD FOR FUELING AN AIRCRAFT**

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(71) Applicant: **JetZero, Inc.**, Long Beach, CA (US)

(52) **U.S. Cl.**

(72) Inventor: **John Vassberg**, Long Beach, CA (US)

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(73) Assignee: **JetZero, Inc.**, Long Beach, CA (US)

(57)

ABSTRACT

(21) Appl. No.: **19/213,576**

In another aspect, a method for fueling an aircraft, the method including storing liquified gas fuel using a fuel tank, wherein the fuel tank is configured to store liquified gas fuel, fueling an aircraft using a fuel line. Fueling the aircraft may additionally include the fuel tank with liquified gas fuel to a desired level, wherein a desired level comprises fuel for a plurality of flights plus reserves. Filling may also include removing the fuel line as a function of the desired level. The method may additionally include venting the fuel tank using a vent line in fluid connection to the fuel tank. The fuel tank may then be prepared for flight as a function of a desired level. Finally, the method includes flying a plurality of flights using the aircraft.

(22) Filed: **May 20, 2025**

Related U.S. Application Data

(63) Continuation of application No. 17/841,153, filed on Jun. 15, 2022, now Pat. No. 12,330,803.

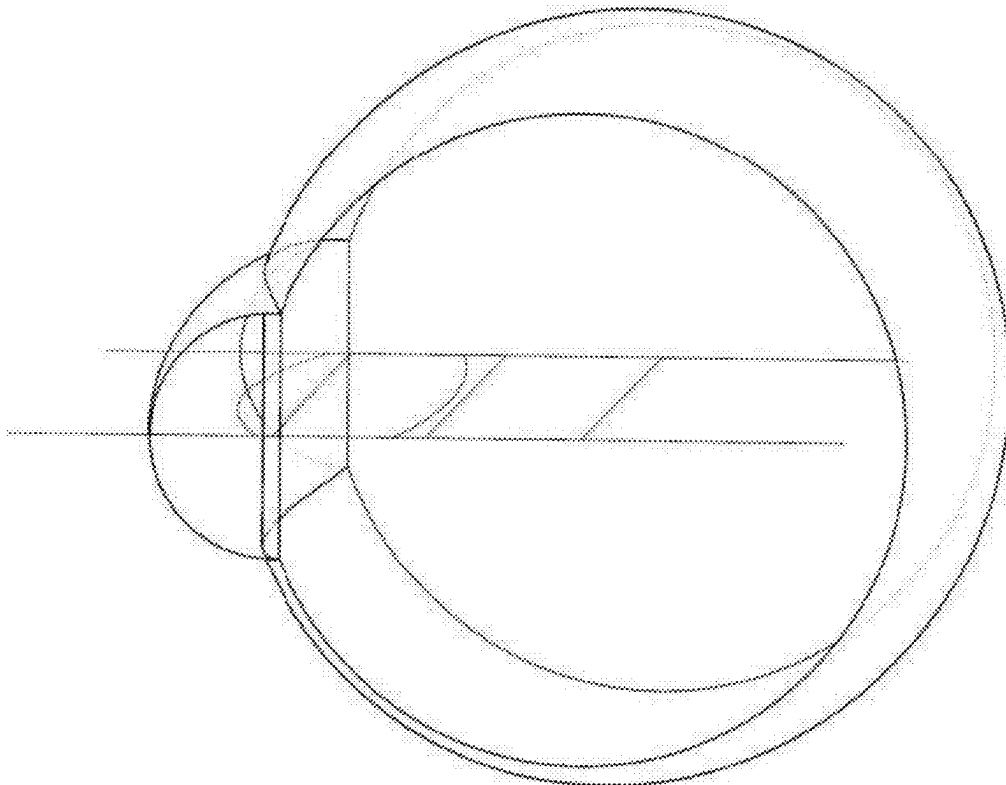
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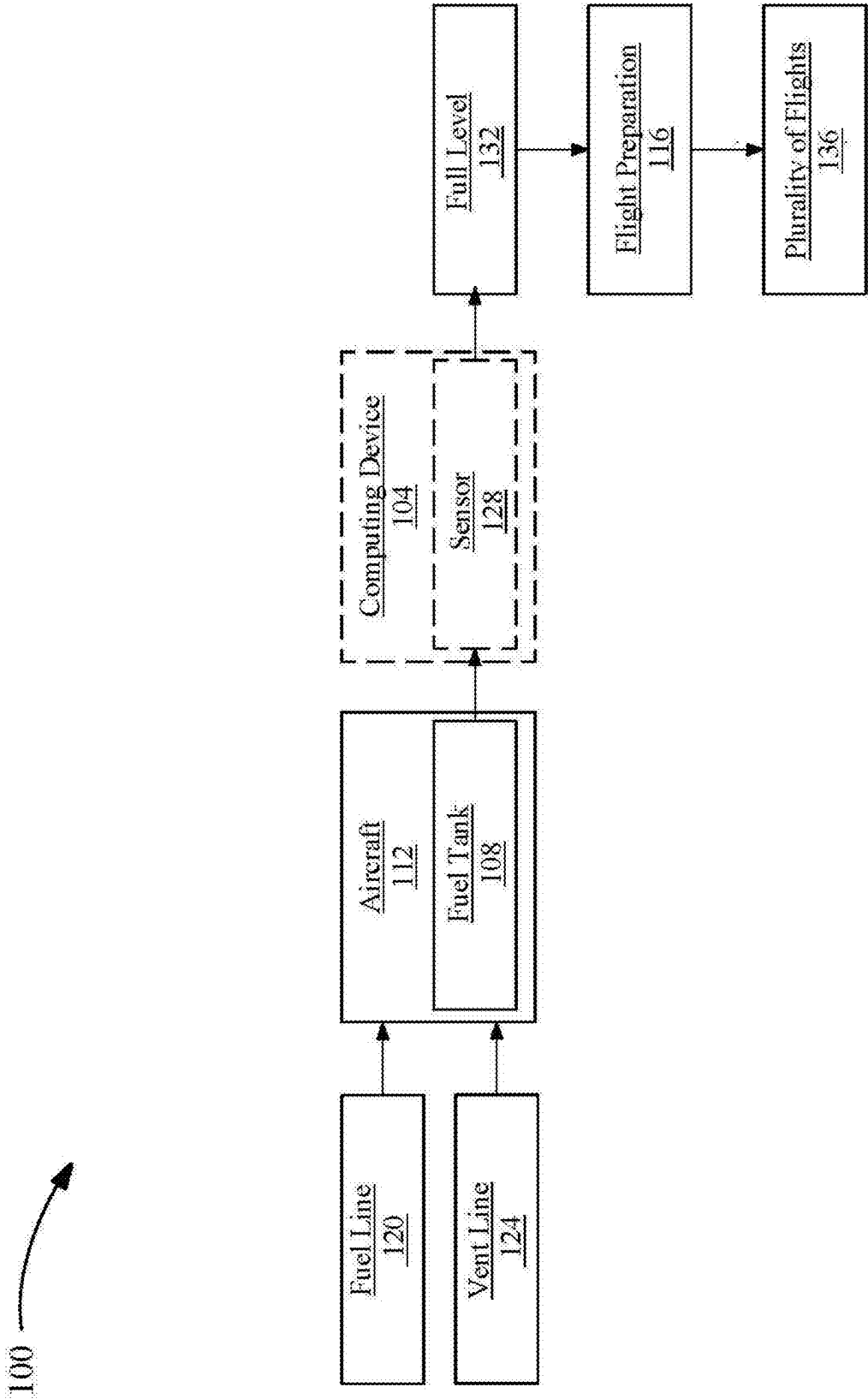
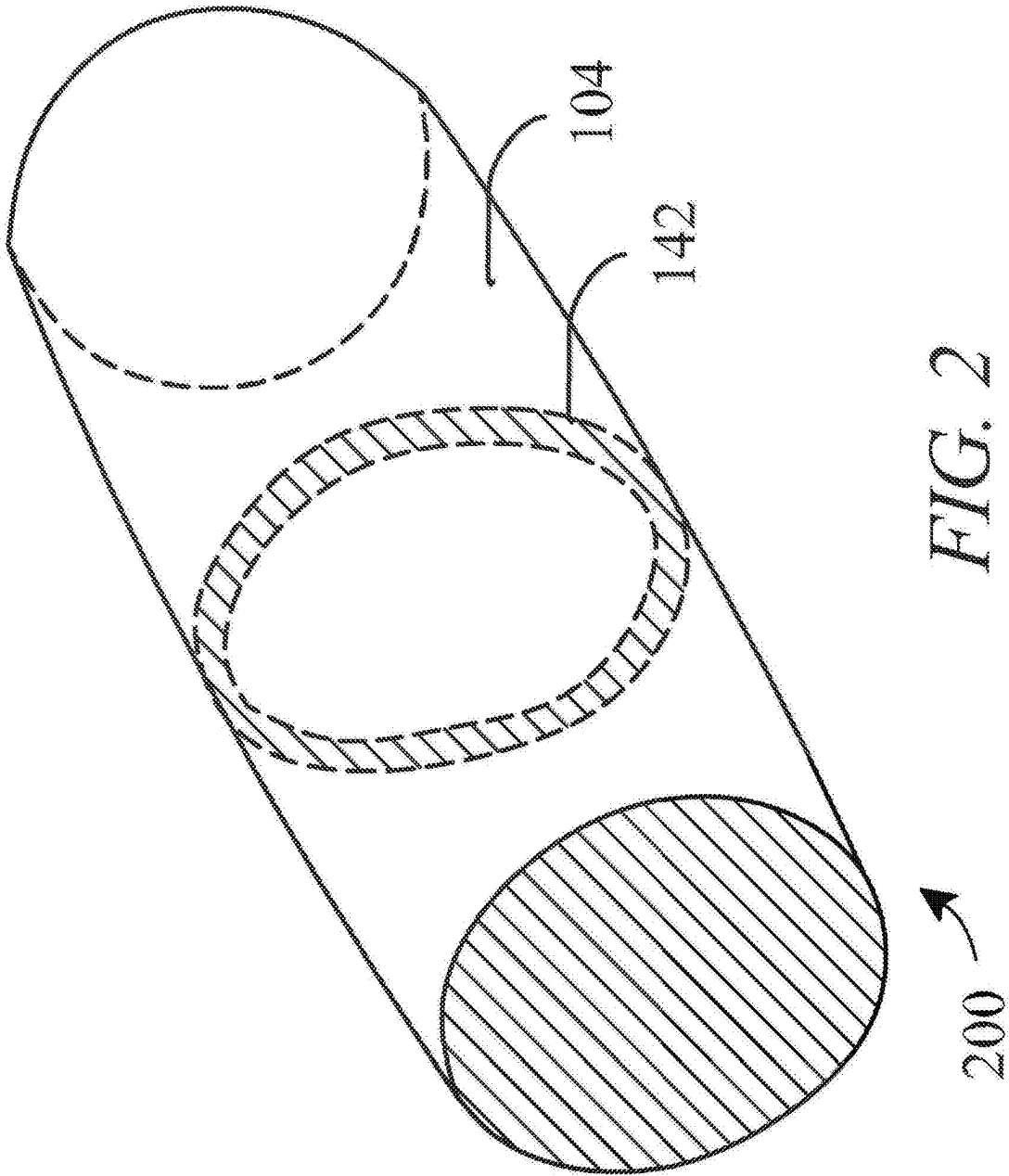
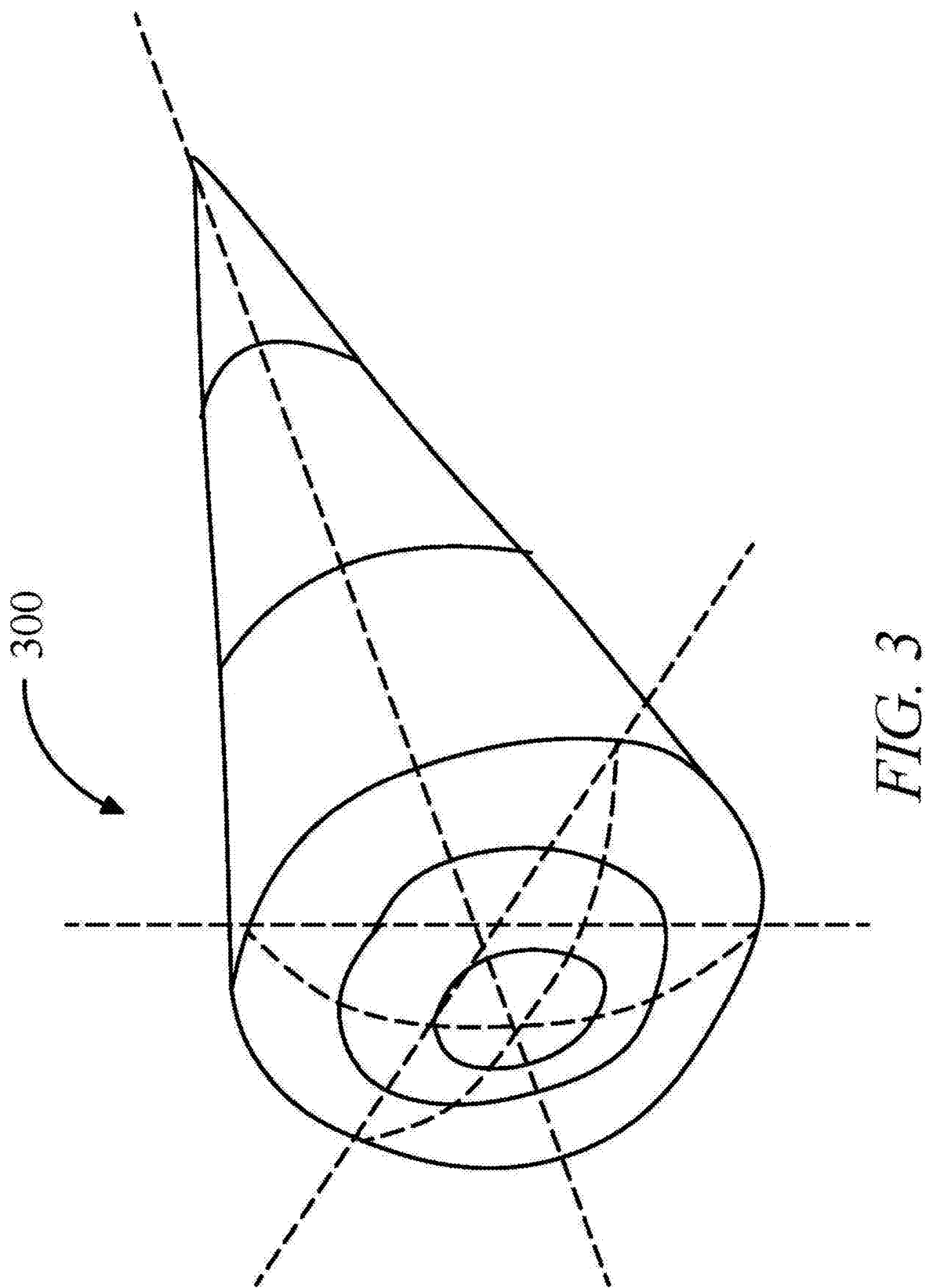


FIG. 1





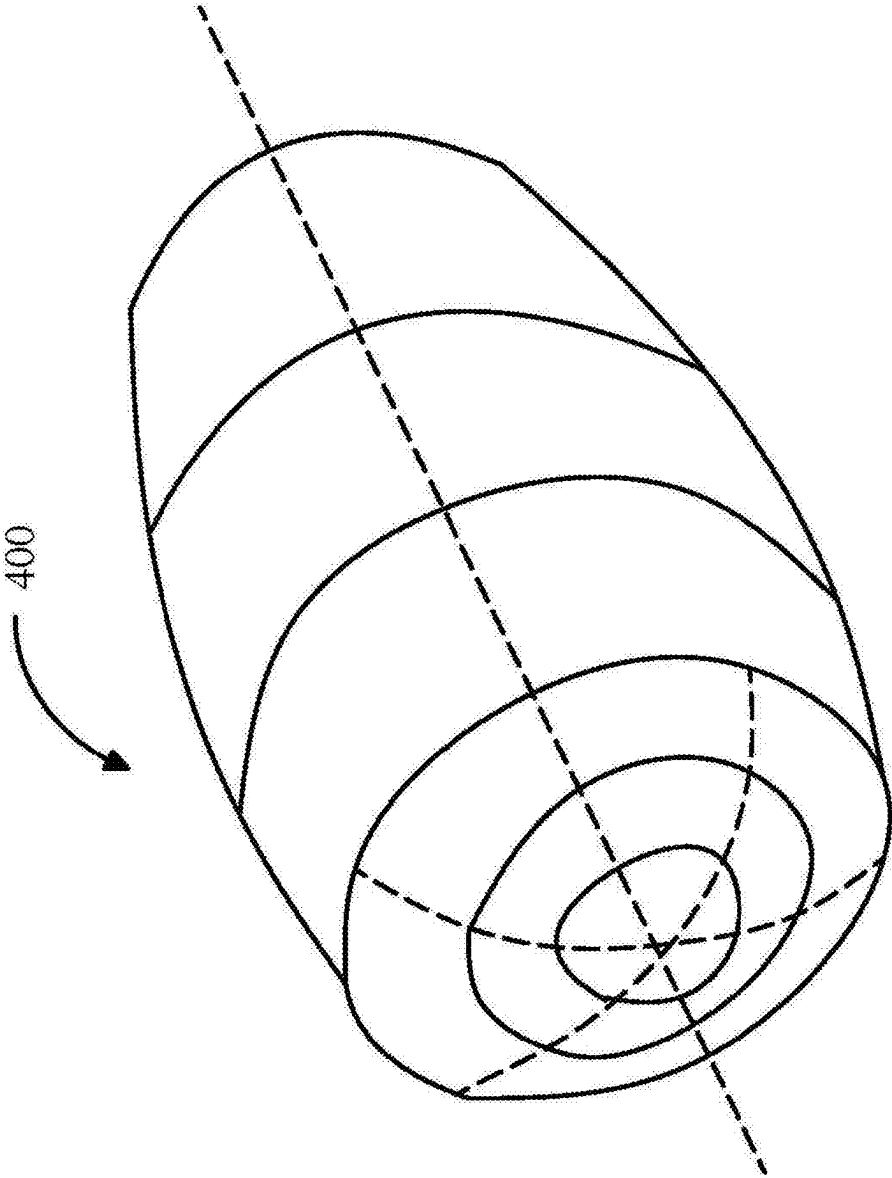


FIG. 4

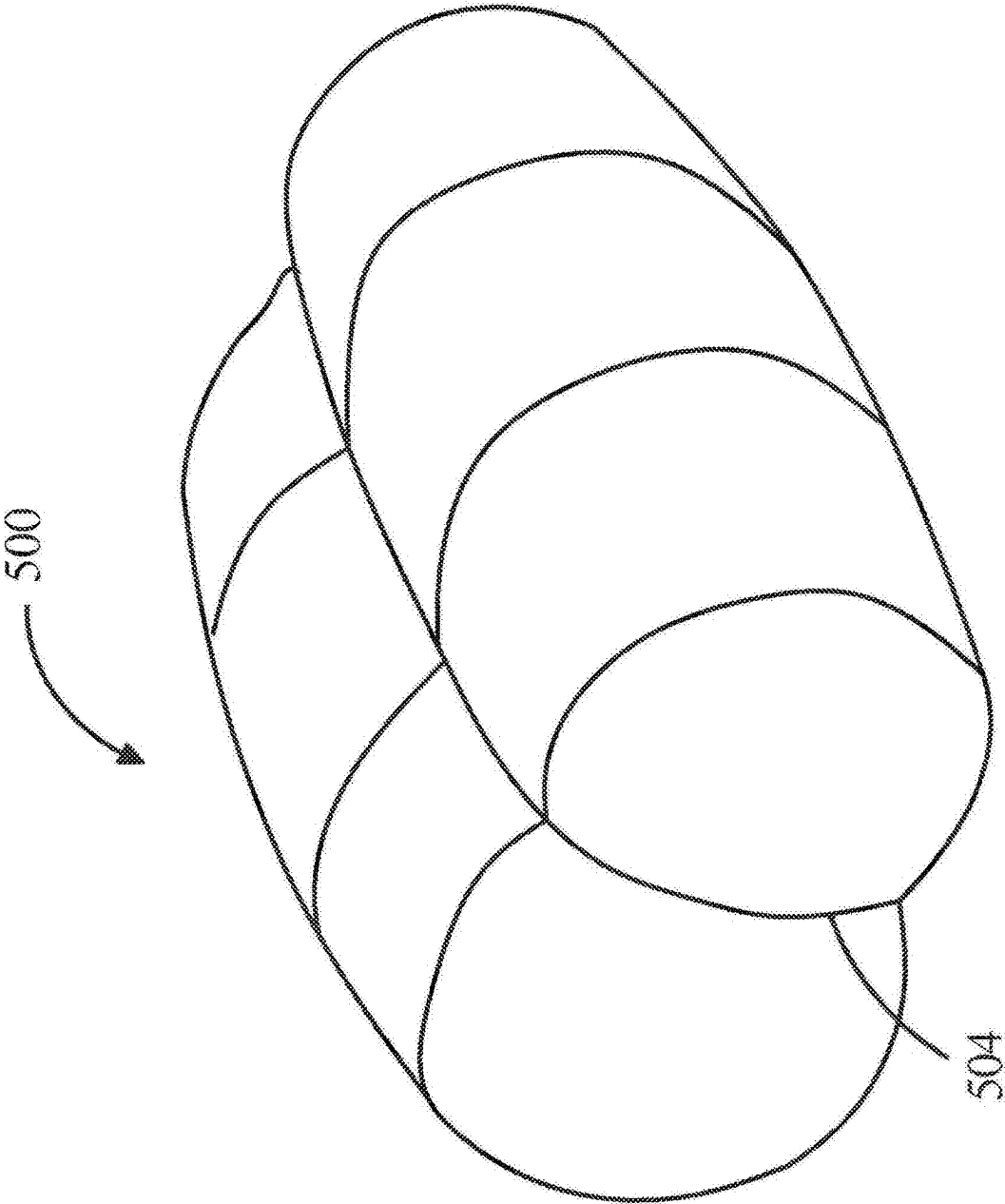


FIG. 5

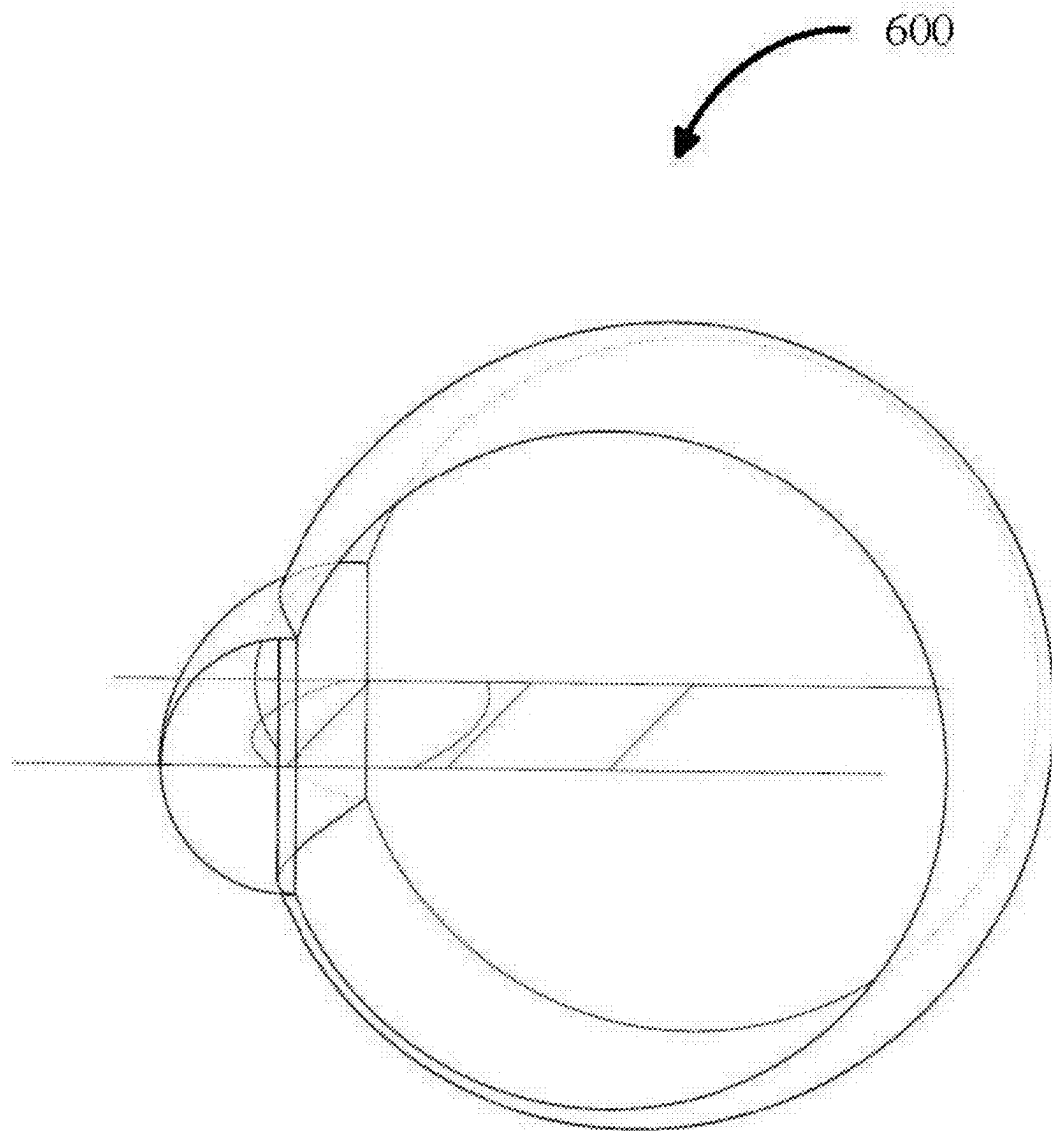


FIG. 6

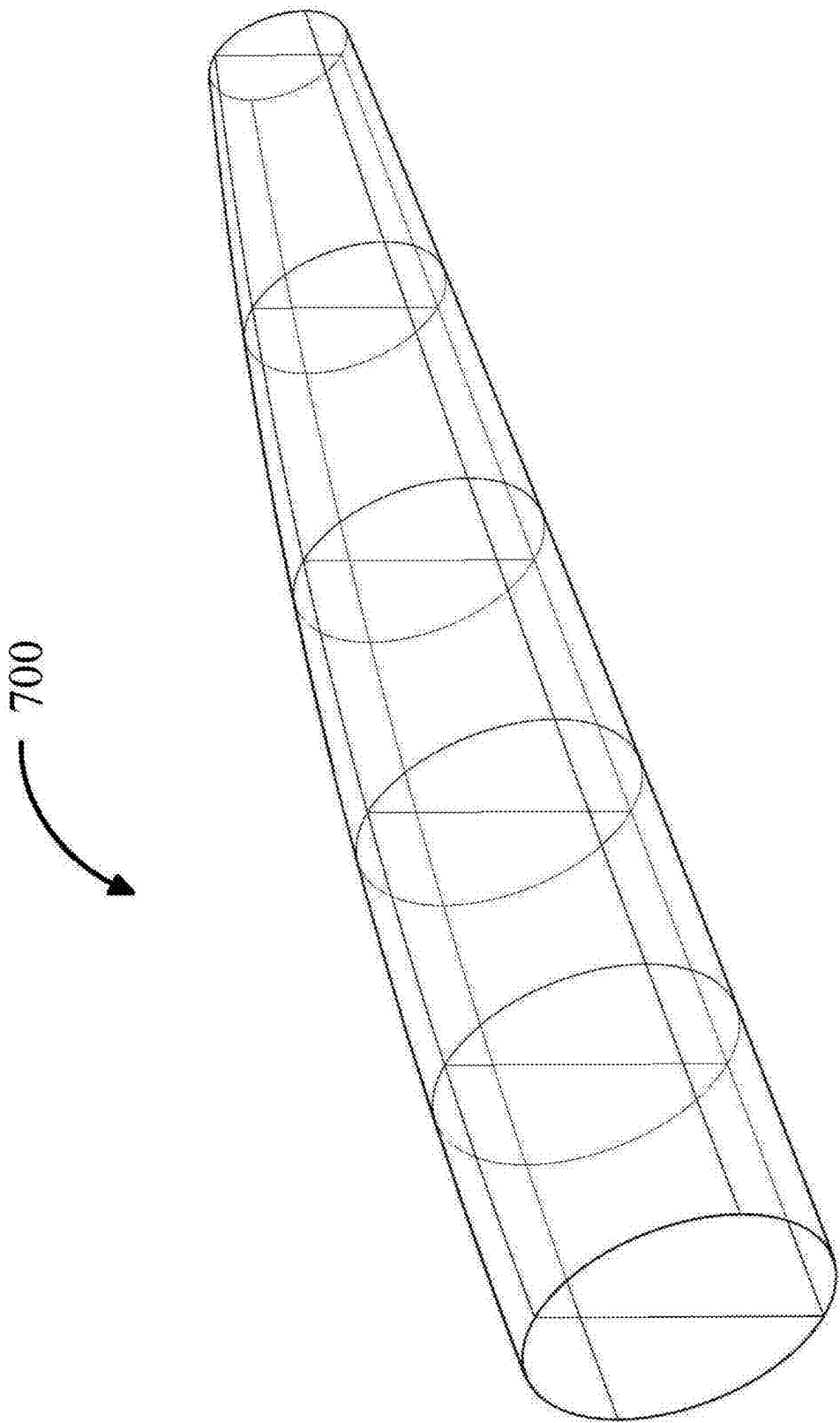


FIG. 7

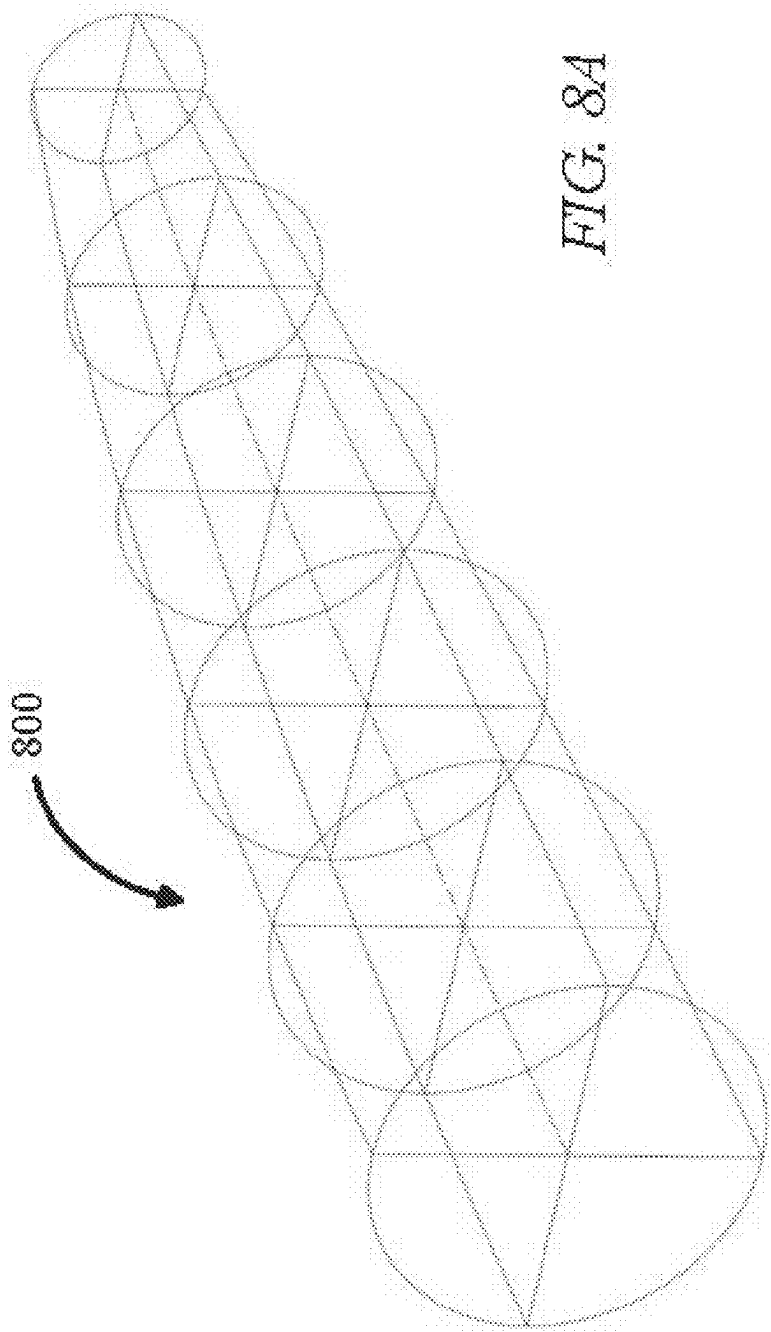


FIG. 8A

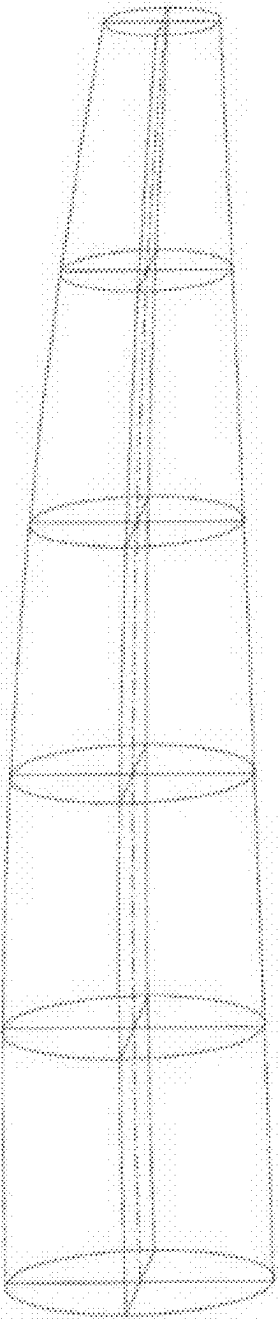


FIG. 8B

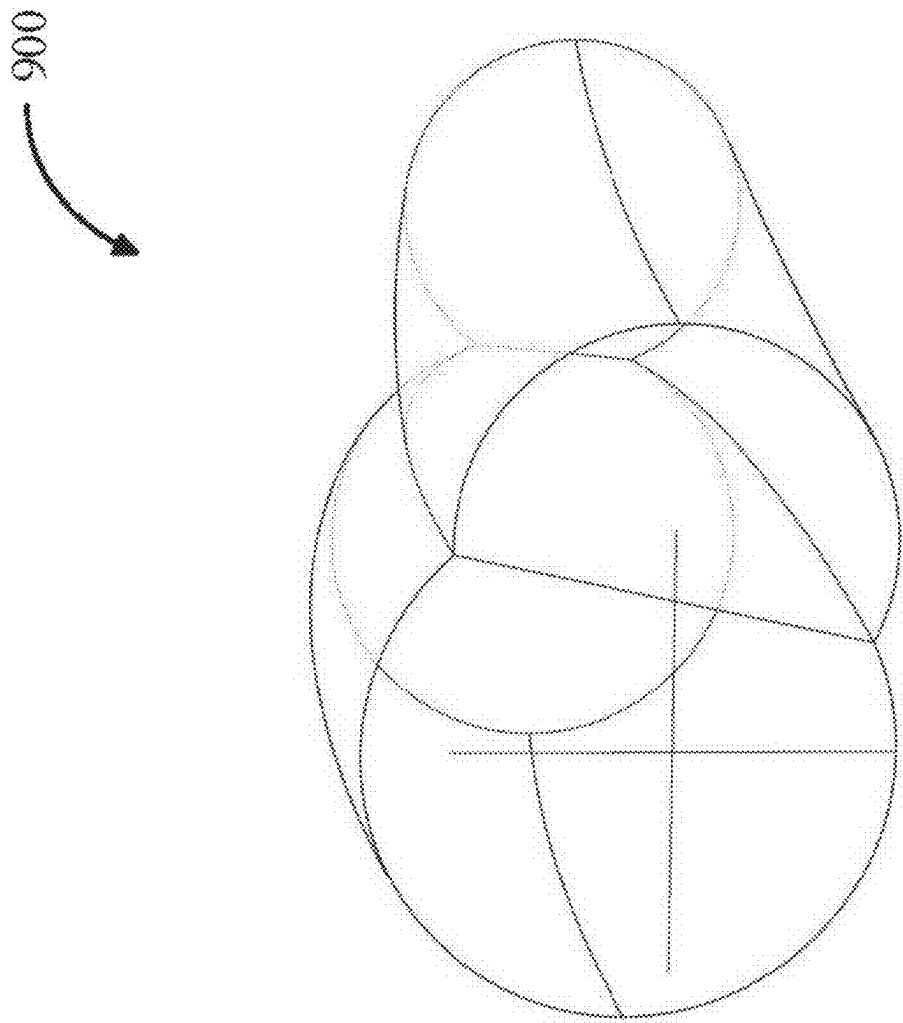


FIG. 9

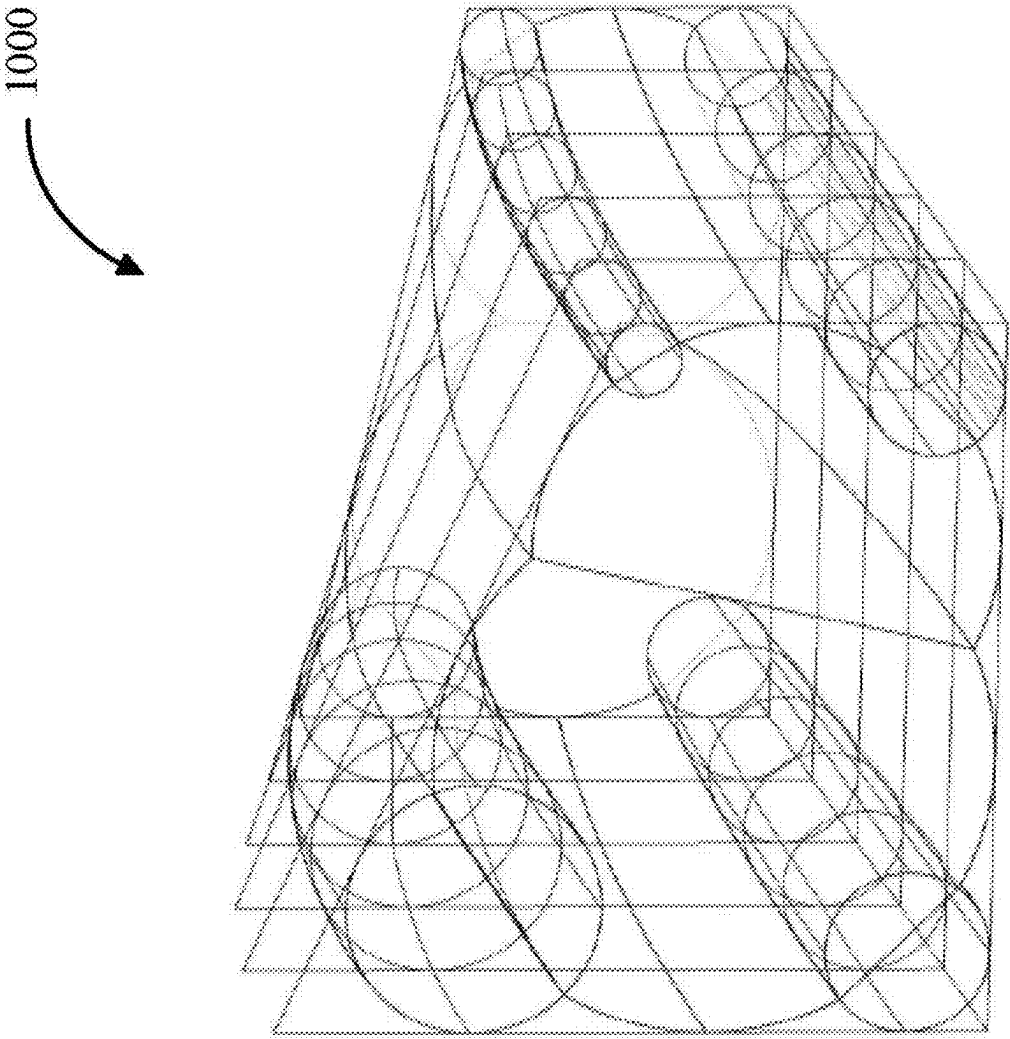


FIG. 10

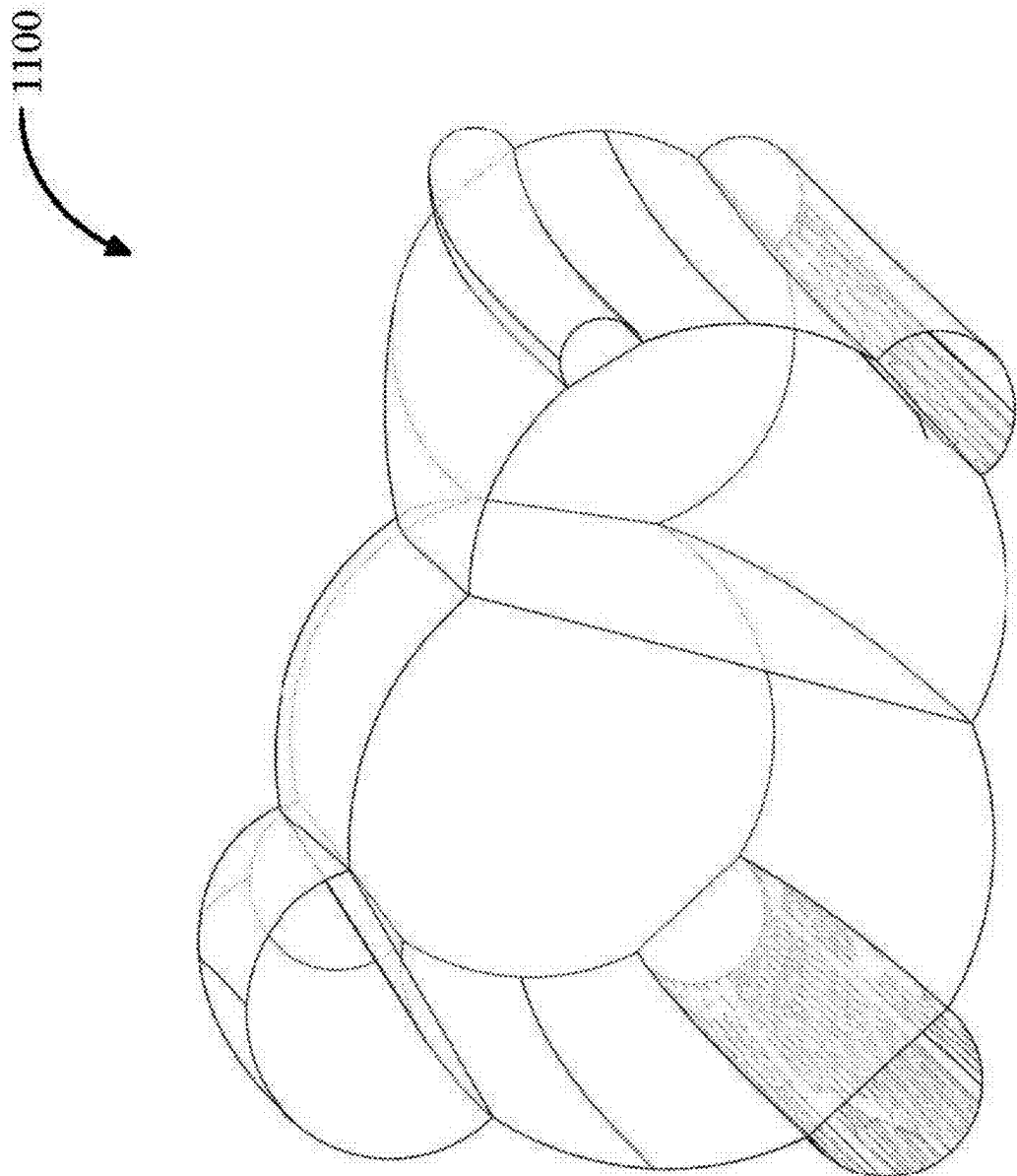


FIG. 11

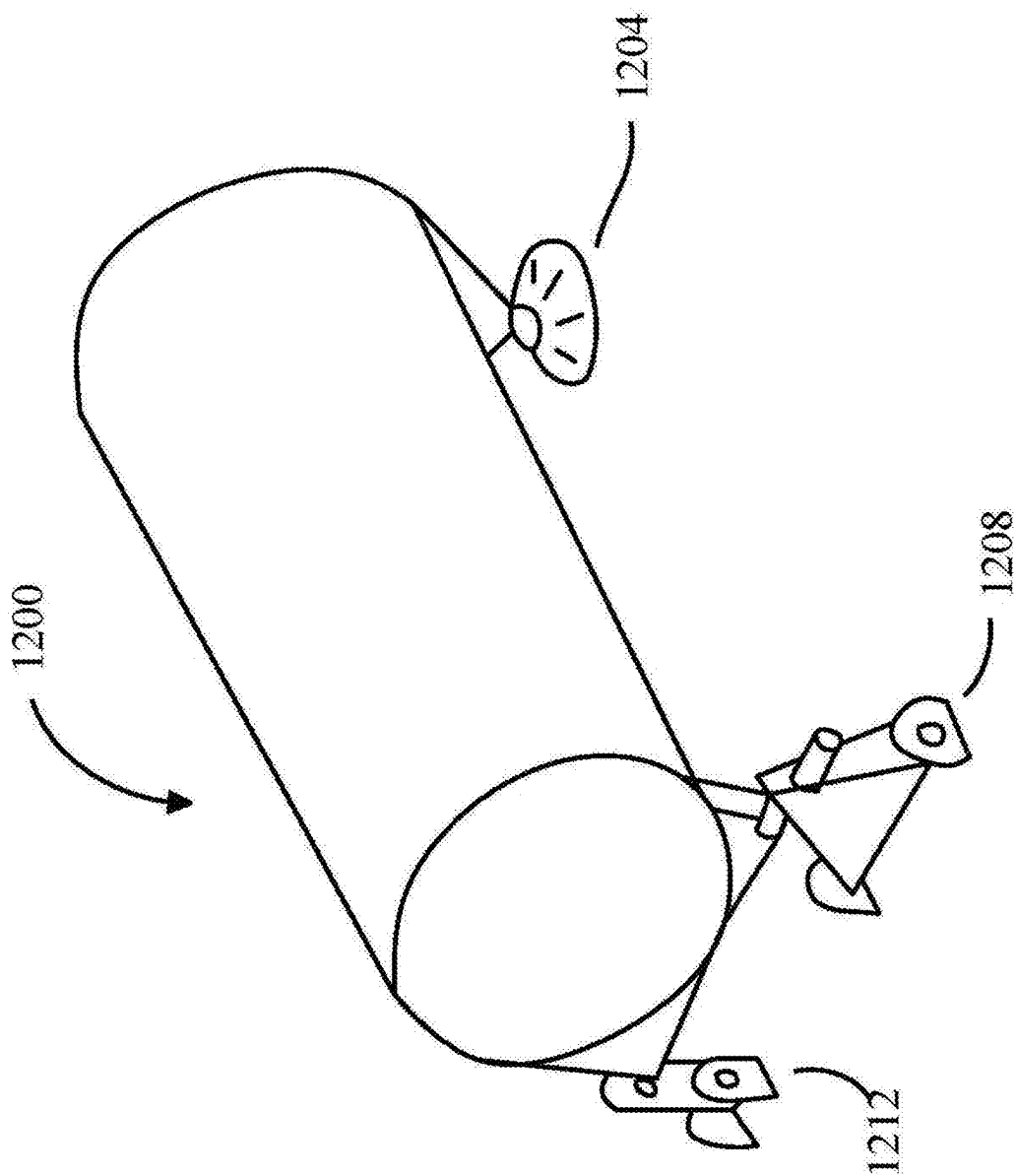


FIG. 12

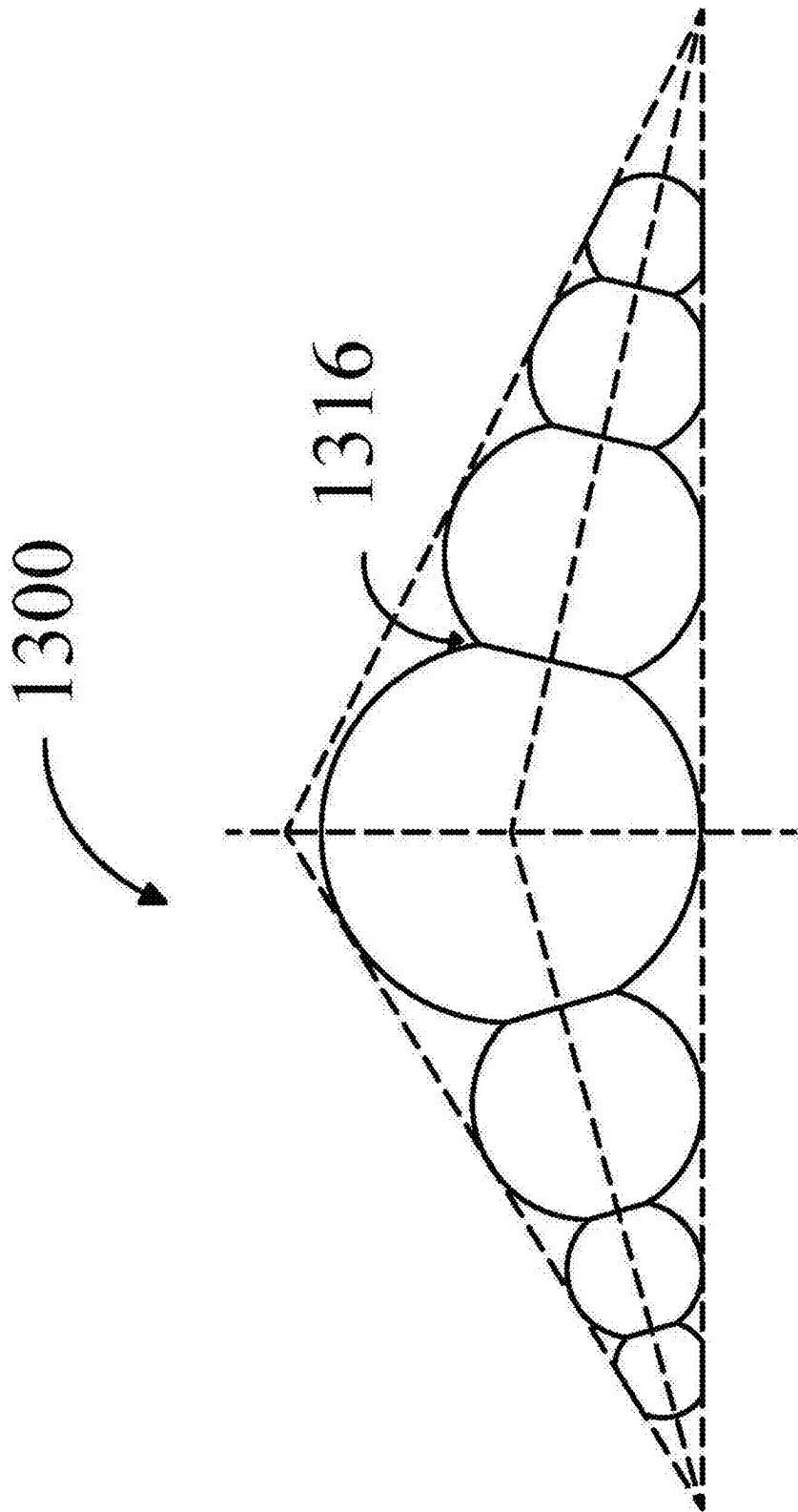
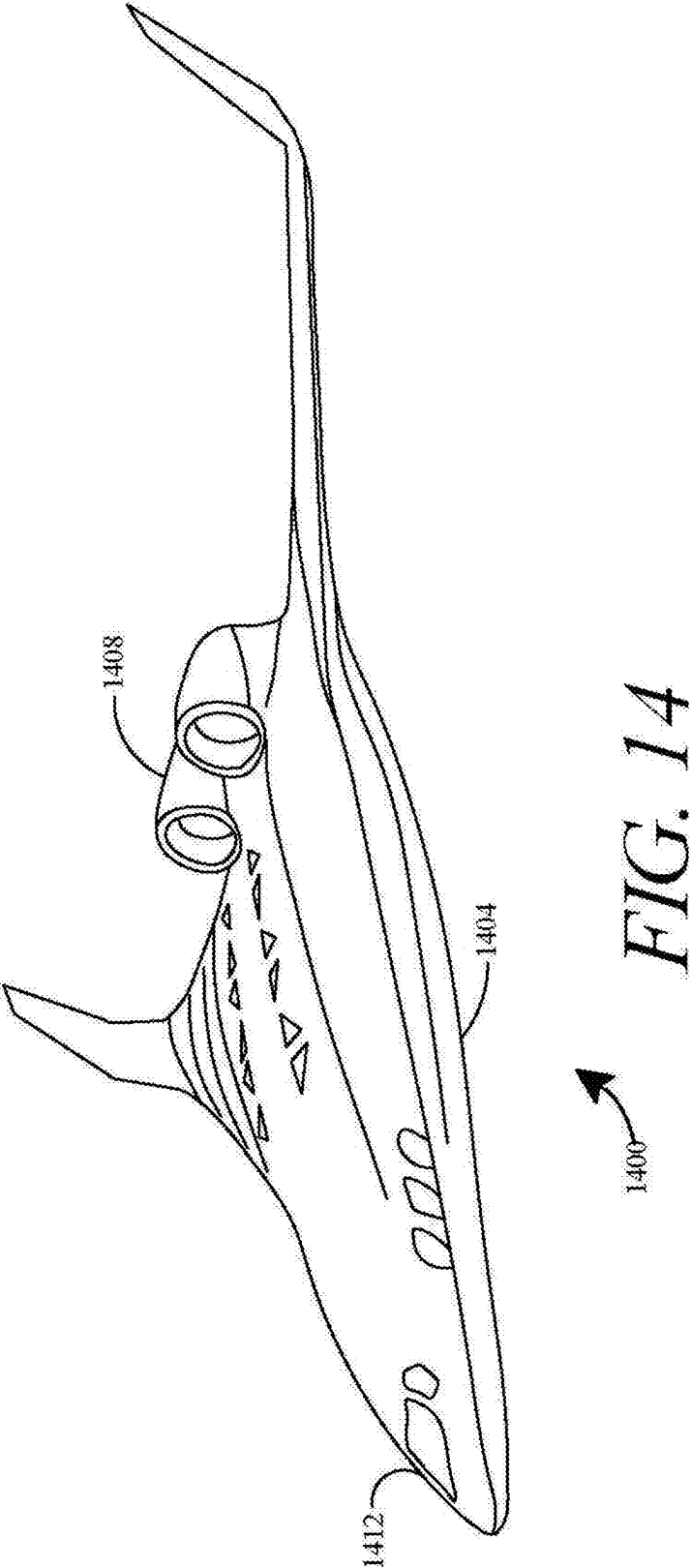


FIG. 13



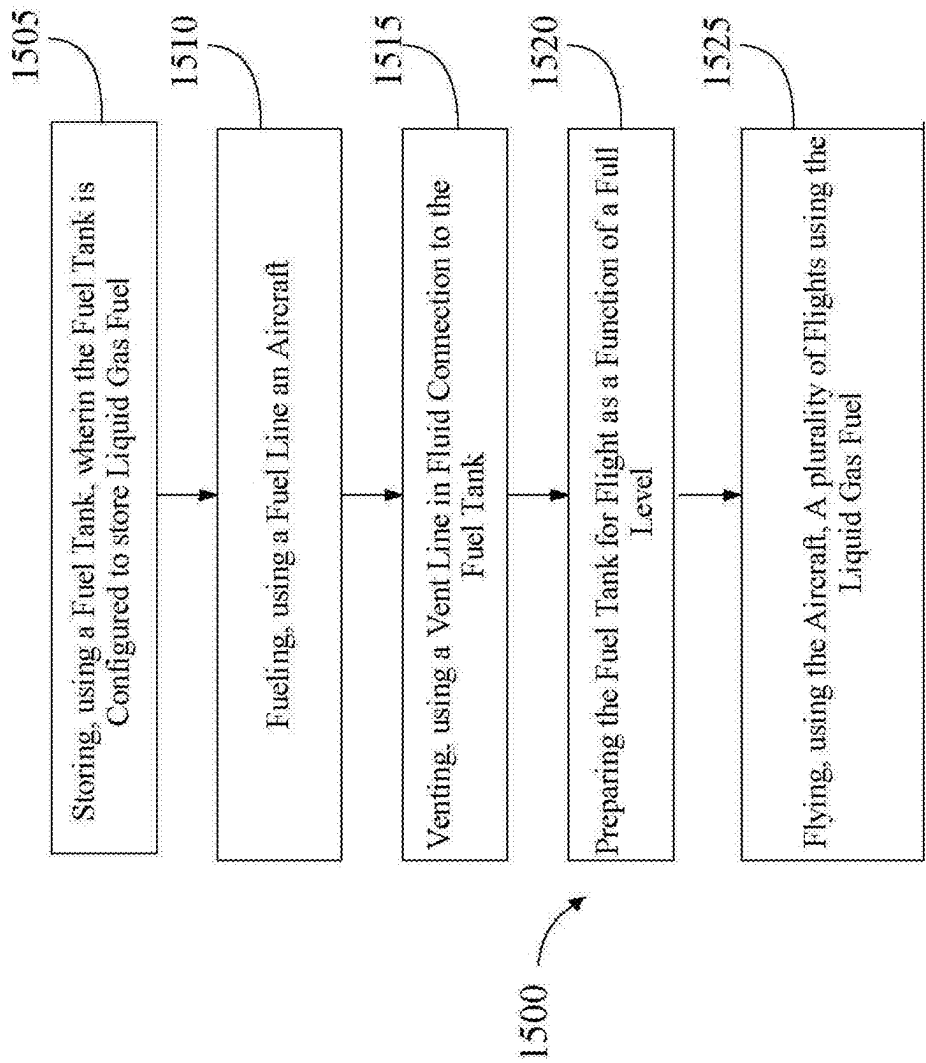


FIG. 15

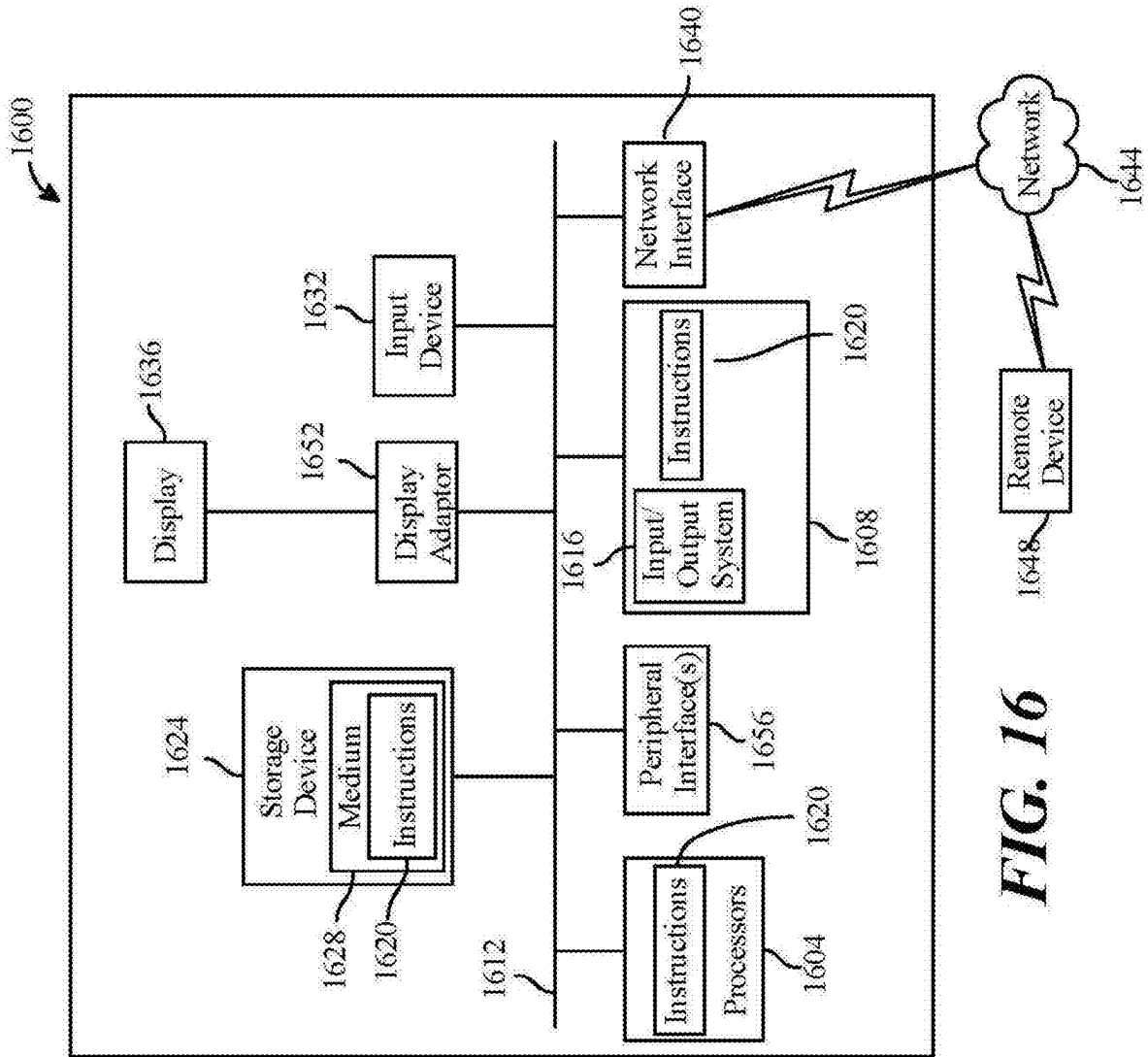


FIG. 16

METHOD FOR FUELING AN AIRCRAFT

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application is a continuation of Non-Provisional application Ser. No. 17/841,153 filed on Jun. 15, 2022, and entitled “METHOD FOR FUELING AN AIRCRAFT,” the entirety of which is incorporated herein by reference.

FIELD OF THE INVENTION

[0002] The present invention generally relates to the field of air travel. In particular, the present invention is directed to a method for fueling an aircraft.

BACKGROUND

[0003] Aircraft are generally designed with strict constraints on fueling time, weight and volume. These constraints apply to all flight components including landing gear. Fueling practices associated with exciting new jetliner aircraft types include additional design requirements that typically require additional components that take less fueling time, space, and weight.

SUMMARY OF THE DISCLOSURE

[0004] In an aspect, a method for fueling an aircraft, may include conditioning a fuel tank of an aircraft, wherein the fuel tank includes a chamber between an inner wall and an outer wall of the fuel tank and clean dry gas is pumped into the chamber to aid in insulating a liquid gas fuel, venting the liquid gas fuel from the chamber, wherein a direction of fuel flow is regulated by a liquid check valve to prevent back flow of the vented liquid gas fuel to the fuel tank, storing, using the fuel tank, liquified gas fuel, fueling the aircraft using a fuel line, wherein fueling includes filling the fuel tank with the liquified gas fuel to a desired level, wherein a sensor connected to the fuel tank detects a plurality of fill data, and preparing the fuel tank for flight as a function of the desired level and the plurality of fill data.

[0005] These and other aspects and features of non-limiting embodiments of the present invention will become apparent to those skilled in the art upon review of the following description of specific non-limiting embodiments of the invention in conjunction with the accompanying drawings.

BRIEF DESCRIPTION OF THE DRAWINGS

[0006] For the purpose of illustrating the invention, the drawings show aspects of one or more embodiments of the invention. However, it should be understood that the present invention is not limited to the precise arrangements and instrumentalities shown in the drawings, wherein:

[0007] FIG. 1 is a block diagram of an exemplary embodiment of a method for fueling an aircraft;

[0008] FIG. 2 is an exemplary embodiment of a tank;

[0009] FIG. 3 is an isometric view of an exemplary embodiment of a conical tank;

[0010] FIG. 4 is an isometric view of an exemplary embodiment of a curved axisymmetric tank;

[0011] FIG. 5 is an isometric view of an exemplary embodiment of a double-curved tank;

[0012] FIG. 6 is a front quarter view of an exemplary embodiment of a dual tank with different diameters;

[0013] FIG. 7 is a quarter side view of an exemplary embodiment of a cambered, tapered tank;

[0014] FIG. 8A illustrates an isometric view of an exemplary embodiment of a dual cambered tank;

[0015] FIG. 8B illustrates a side quarter view of an exemplary embodiment of a dual cambered tank;

[0016] FIG. 9 shows a quarter front view of an exemplary embodiment of a double tank;

[0017] FIG. 10 is a front quarter view of an exemplary embodiment of a multi-lobe tank without trimming;

[0018] FIG. 11 is a front quarter view of an exemplary embodiment of a multi-lobe tank trimmed with septa;

[0019] FIG. 12 illustrates an isometric view of an exemplary embodiment of a tank with a plurality of tank support links;

[0020] FIG. 13 illustrates an exemplary embodiment of a cross-sectional view of multi-lobe tanks;

[0021] FIG. 14 is an exemplary embodiment of a blended wing body aircraft;

[0022] FIG. 15 is a block diagram for a method of use for fueling an aircraft; and

[0023] FIG. 16 is a block diagram of a computing system that can be used to implement.

[0024] The drawings are not necessarily to scale and may be illustrated by phantom lines, diagrammatic representations, and fragmentary views. In certain instances, details that are not necessary for an understanding of the embodiments or that render other details difficult to perceive may have been omitted.

DETAILED DESCRIPTION

[0025] At a high level, aspects of the present disclosure are directed to systems and methods for fueling an aircraft. The method includes storing liquified gas fuel using a fuel tank, wherein the fuel tank is configured to store liquified gas fuel, fueling an aircraft using a fuel line. Fueling the aircraft may additionally include the fuel tank with liquified gas fuel to a desired level, wherein a desired level comprises fuel for a plurality of flights plus reserves. Filling may also include removing the fuel line as a function of the desired level. The method may additionally include venting the fuel tank using a vent line in fluid connection to the fuel tank. The fuel tank may then be prepared for flight as a function of a desired level. Finally, the method includes flying a plurality of flights using the aircraft.

[0026] In the following description, for the purposes of explanation, numerous specific details are set forth in order to provide a thorough understanding of the present invention. For purposes of description herein, relating terms, including “upper,” “lower,” “left,” “rear,” “right,” “front,” “vertical,” “horizontal,” and derivatives thereof relate to embodiments oriented as shown for exemplary purposes in FIG. 14. Furthermore, there is no intention to be bound by any expressed or implied theory presented in this disclosure.

[0027] Referring now to FIG. 1, an exemplary embodiment of a method 100 for fueling an aircraft is illustrated. System includes a computing device 104, computing device 104 may include any computing device as described in this disclosure, including without limitation a microcontroller, microprocessor, digital signal processor (DSP) and/or system on a chip (SoC) as described in this disclosure. Computing device may include, be included in, and/or commu-

nicate with a mobile device such as a mobile telephone or smartphone, computing device 104 may include a single computing device operating independently, or may include two or more computing devices operating in concert, in parallel, sequentially or the like; two or more computing devices may be included together in a single computing device or in two or more computing devices, computing device 104 may interface or communicate with one or more additional devices as described below in further detail via a network interface device. Network interface device may be utilized for connecting computing device 104 to one or more of a variety of networks, and one or more devices. Examples of a network interface device include, but are not limited to, a network interface card (e.g., a mobile network interface card, a LAN card), a modem, and any combination thereof. Examples of a network include, but are not limited to, a wide area network (e.g., the Internet, an enterprise network), a local area network (e.g., a network associated with an office, a building, a campus or other relatively small geographic space), a telephone network, a data network associated with a telephone/voice provider (e.g., a mobile communications provider data and/or voice network), a direct connection between two computing devices, and any combinations thereof. A network may employ a wired and/or a wireless mode of communication. In general, any network topology may be used. Information (e.g., data, software etc.) may be communicated to and/or from a computer and/or a computing device, computing device 104 may include but is not limited to, for example, a computing device or cluster of computing devices in a first location and a second computing device or cluster of computing devices in a second location, computing device 104 may include one or more computing devices dedicated to data storage, security, distribution of traffic for load balancing, and the like, computing device 104 may distribute one or more computing tasks as described below across a plurality of computing devices of computing device, which may operate in parallel, in series, redundantly, or in any other manner used for distribution of tasks or memory between computing devices, computing device 104 may be implemented using a “shared nothing” architecture in which data is cached at the worker, in an embodiment, this may enable scalability of method 100 and/or computing device.

[0028] With continued reference to FIG. 1, computing device 104 may be designed and/or configured to perform any method, method step, or sequence of method steps in any embodiment described in this disclosure, in any order and with any degree of repetition. For instance, computing device 104 may be configured to perform a single step or sequence repeatedly until a desired or commanded outcome is achieved; repetition of a step or a sequence of steps may be performed iteratively and/or recursively using outputs of previous repetitions as inputs to subsequent repetitions, aggregating inputs and/or outputs of repetitions to produce an aggregate result, reduction or decrement of one or more variables such as global variables, and/or division of a larger processing task into a set of iteratively addressed smaller processing tasks, computing device 104 may perform any step or sequence of steps as described in this disclosure in parallel, such as simultaneously and/or substantially simultaneously performing a step two or more times using two or more parallel threads, processor cores, or the like; division of tasks between parallel threads and/or processes may be performed according to any protocol suitable for division of

tasks between iterations. Persons skilled in the art, upon reviewing the entirety of this disclosure, will be aware of various ways in which steps, sequences of steps, processing tasks, and/or data may be subdivided, shared, or otherwise dealt with using iteration, recursion, and/or parallel processing.

[0029] Still referring to FIG. 1, as used in the current disclosure, a “fuel tank” is a container specifically designed to hold fuel. A fuel tank may also be referred to as a “tank.” A fuel tank 108 of a plurality of fuel tanks may include one or more compartments to store fuel in. A fuel tank 108 of a plurality of fuel tanks may be a part of fuel delivery system for an engine, in which the fuel may be stored inside a fuel tank 108 and then propelled or released into an engine, such as without limitation a combustion engine. A fuel tanks may be configured to be removably attached to an aircraft 112. A fuel tank 108 may be configured to contain enough fuel for multiple flights or trips. A fuel tank 108 may also be configured to hold an amount of fuel that is predetermined by the flight plan of the aircraft. In some cases, fueling may be performed at a slow rate over a long period of time. For example, in some cases, fueling may take one hour, two hours, three hours, five hours, or eight hours. In some cases, aircraft may be fueled overnight, or when substantially no flights are scheduled for a long period of time, i.e., not between change-overs between flights.

[0030] With continued reference to FIG. 1, method 100 may include a fuel tank 108. In an embodiment, a fuel tank 108 stores fuel to power aircraft 112. A fuel tank 108 may be permanently attached to aircraft 112. As used in this disclosure, a tank may be “permanently attached” when it is configured to not be removed during ordinary use. For example, a tank permanently attached to aircraft may be removed during maintenance or overhaul but is otherwise a permanent flight component of the aircraft. A fuel tank 108 of a plurality of fuel tanks may include one or more compartments to store fuel in. A fuel tank 108 of a plurality of fuel tanks may be a part of fuel delivery system for an engine, in which the fuel may be stored inside a fuel tank and then pumped or released into an engine, such as without limitation a combustion engine.

[0031] With continued reference to FIG. 1, fuel tank 108 may be a pressure vessel. A “pressure vessel” is a container configured to hold fluids at a pressure that differ from an ambient pressure. Pressure vessel may be configured to be pressurized in order to allow flow of gaseous hydrogen from a fuel tank 108, for example without a need to use a pump. In an embodiment, but without limitation, a fuel tank 108 may act as a pressure vessel to store the fuel at a high pressures above 5 psig, 15 psig, 50 psig, or the like. A fuel tank 108 may be made of any material able to withstand such high pressure, such as but without limitation, aluminum, carbon fiber, composite materials, or the like. Furthermore, a fuel tank 108 may further include an inner wall and an outer wall. As used herein, an “inner wall” is the inner barrier of a tank that is in contact with the fuel. As used herein, an “outer wall” is the outer barrier of a tank that is exposed to the outside. There may be insulation between the outer and inner wall. A fuel tank 108 may also include safety valves, closures, vessel threads, or any other features that can be found on fuel tanks.

[0032] With continued reference to FIG. 1, Fuel tank 108 may further have a tank geometry. “Tank geometry” refers to an overall shape and arrangement of fuel tank 108. Fuel

tank **108** may have a multi-lobe geometry, such that the multi-lobe geometry includes one or more curvatures. As used herein, “multi-lobe geometry” refers to a shape include multiple curvatures or “lobes.” A “lobe” as used herein, is a curved section of the multi-lobe geometry. Multi-lobe geometry may have multiple lobes, wherein the start of one lobe is marked by a discontinuation in the curvature. A discontinuation in the curvature may occur when the curvature of a tank switches between two non-adjacent values instantaneously or substantially instantaneously. The curvature of the tank may switch substantially instantaneously when there is a weld or other fastener or a division such as a septum. In other embodiments, the curvature may switch substantially instantaneously, when, for example, two lobes of a tank meet, but the intersection has been rounded or otherwise altered in order to, among other things, reduce stress concentrations at the intersection point. For example, the fuel tank **108** may have one or more surfaces with spherical and/or cylindrical shapes. Each lobe of the multi-lobe tank geometry may be configured to have a different radius. Fuel tank **108** may include variable diameter and length. A lobe may be defined as a single sphere or cylinder of the plurality of spheres and/or cylinders. According to some embodiments, tank geometry for a blended wing body aircraft may be driven by at least five objectives: (1) to provide as much fuel volume as possible while using little payload floor space, (2) to provide a fuel tank shape that resists pressure and that is lightweight, (3) to provide a fuel tank shape that can be insulated between outer wall and inner wall, (4) to provide a fuel center of gravity that is not widely misaligned with the aircraft’s center of gravity, and (5) to provide fuel tanks that are compatible with a passenger cabin. In this disclosure, a “fuel center of gravity” is the center of gravity of the fuel inside fuel tank **108**. In some cases, fuel center of gravity may affect overall aircraft center of gravity. Fuel center of gravity may be associated with a volume within airplane that has sufficient volume to store a practical quantity of fuel in discrete tanks. Fuel tank **108** may be arranged inside blended wing body as a function of its fuel center of gravity in relation to center of gravity of aircraft **112**. Tank geometry is further discussed herein with reference to FIGS. 2-13.

[0033] With continued reference to FIG. 1, method **100** includes an aircraft **112**. Aircraft **112** may include an exemplary blended wing aircraft. Aircraft **112** may include a blended wing body **1404** of FIG. 14. For the purposes of this disclosure, a “blended wing body aircraft” is an aircraft having a blended wing body. As used in this disclosure, A “blended wing body” (BWB), also known as a “blended body” or a “hybrid wing body” (HWB), is a fixed-wing aircraft body having no clear or abrupt demarcation between wings and a main body of the aircraft along a leading edge of the aircraft. Blended wing body aircraft is discussed in further detail in FIG. 14.

[0034] With continued reference to FIG. 1, aircraft **112** includes a fuel tank **108**, which may further include a liquified gas fuel, and may include at least a vent, and an insulation. Fuel tank **108** may be a permanent tank. A “permanent tank” is a tank that is permanently attached to aircraft **112**. A permanent tank may be added to an existing airplane or attached during the airplane’s construction. In either case, the intention of a permanent tank may be to leave fuel tank **108** in place for an extended period. A permanent tank may integrate fuel tank **108** and BWB **1404** structure so

that the BWB **1404** supports fuel tank **108**, and/or fuel tank **108** supports BWB **1404**. As a non-limiting example, Fuel tank **108** may support BWB **1404** by assisting the airframe of the BWB **1404** in resisting pressurization and carrying shear that would otherwise be carried by the ribs of the BWB **1404**. In some embodiments, curved surfaces of fuel tank **108** may not be well suited to assisting structural surfaces of BWB **1404**. However, in some embodiments, septa of multi-lobe tanks may be well-suited to support the structure of aircraft **112**. As used herein, “septa” are partitions between objects, such as two fuel tanks. The plurality of fuel tanks may be divided by septa such that there is a septum between each fuel tank. In an embodiment, the septa may extend from a lower outer mold line to an upper outer mold line of a fuel tank **108**. A top and bottom of fuel tank **108** may be closed out by spherical end caps inset from the outer mold line to provide room for the outer mold line skin’s supporting structure. This may assist aircraft **112** in resisting pressurization and simultaneously carrying the shear stress that is otherwise carried by the ribs of the airframe of aircraft **112**.

[0035] Still referring to FIG. 1, fuel tank **108** is filled with a liquified gas fuel. Liquified gas fuel may include a liquid hydrogen fuel. As used in this disclosure, a “liquified gas fuel” is a fuel that at standard atmospheric conditions or when utilized (e.g., combusted) is gas and is stored as a fuel. Liquified gas fuels include without limitation liquid hydrogen, propane, and liquified natural gas. Liquified gas fuel may be stored as a compressed gas or in liquid form. Liquid hydrogen fuel may be used for airliner-scale systems, liquid hydrogen fuel may a higher density than gaseous hydrogen; even so, about four times as much liquified gas fuel may be needed compared to the volume of Jet-A fuel needed. Additionally, liquified gas fuel may also allow for reduced tank pressure and tank weight. Liquified gas fuel may be kept at extremely low, cold temperatures, for example below its critical point of 33 Kelvin. Gas turbine engines, such as combustion engines, may operate on gaseous hydrogen and may transition liquified gas fuel into a gas before consumption; gaseous hydrogen may be supplied to the engine at high pressure, but typically no fuel pump is used. In some embodiments, the gas may squirt into a combustion chamber due to compressor stages. Hydrogen vapor from tank ullage may be combined with the hydrogen vapor from an output of a fuel heater. In some cases, this combined channel may then lead to engine for combustion. When boiled, a rate of boil-off for liquified gas fuel may be determined as a function of heat energy applied to liquified gas fuel. Boil-off for liquified gas fuel can be selected, but fuel tank **108** may be insulated with an insulation to control application of heat to tank contents. Fuel tank **108** may also be refrigerated to remove heat from liquified gas fuel. The degree of insulation may be selected to provide a desired rate of boil-off. Insulation is further described below. The selected rate of boil-off for liquified gas fuel may generally be less than the rate at which liquified gas fuel is consumed by the engines and possibly other aircraft systems, such as without limitation an APU. Excess boil-off may be dumped overboard or outside of aircraft through a vent.

[0036] Continuing to refer to FIG. 1, fuel tank **108** may include at least a vent. As used in this disclosure, a “vent” is an opening and/or aperture configured to allow one or more fluids to pass. In an embodiment, at least a vent may be configured to vent gasses from fuel tank **108**. In some

embodiments, these gasses may include gaseous hydrogen. Gaseous hydrogen may result from boil-off of liquified gas fuel as the fuel warms. In an embodiment, and without limitation, at least a vent may be configured to vent boil-off from fuel tank 108. In some cases, at least a vent may include a check valve. As used in this disclosure, a “check valve” is a valve that permits flow of a fluid only in certain (e.g., one) directions. In some cases check valve may be configured to allow flow of fluids substantially only away from fuel tank 108 while preventing back flow of vented fluid to fuel tank 108. At least a vent may also include a pressure regulator. A “pressure regulator” is a type of valve that controls the pressure of a fluid. Venting gasses from a fuel tank 108 prevents over-pressurizing or other events that may cause catastrophic damage or harm. It may also desirable, when aircraft 112 is grounded, to connect a system of lines and tanks to at least a vent to collect the boiled-off hydrogen. In some cases, the collected gaseous hydrogen or fuel gasses can be compressed by a pump into storage tanks and then cooled to liquid temperatures for reuse as aircraft fuel. In embodiments a vent may be in fluid connection to both a fuel tank 108 and an interface port. In other embodiments, a vent may be attached to an interface port and a vent line 124.

[0037] Still referring to FIG. 1, fuel tank 108 may include insulation. In this disclosure, “insulation” is a component configured to reduce heat transfer. Insulation may be used to reduce thermal transfer to liquified gas fuel inside of a fuel tank 108. Heat may be transferred to fuel tank 108 by at least two means: conduction and radiation. To reduce conduction, insulation may include a vacuum to separate an interior volume of fuel tank 108 from an exterior with an evacuated vessel. Another means to reduce conduction in fuel tank 108 may be an insulating material that inhibits conduction. Insulating materials include fiberglass wool, plastic or ceramic foam, aerogel, and other materials. Heat flow through an insulator can be inversely proportional to its thickness. For example, an insulator that is twice as thick may conduct heat at half the rate. Heat transfer by radiation may be reduced by reflective coatings. Reflective coatings may be located proximal surfaces of fuel tank 108 and/or on intermediate materials within insulation. For example, a dewar may be coated with a mirror-like material that reflects heat radiation; many thermos bottles are silvered for this reason. A “dewar” is double-walled flask of metal or silvered glass with a vacuum between the walls and configured to hold a liquid. Another means to reduce radiation may be to sandwich multiple layers of thin reflective foil within an insulative material such as foam. Additionally, a modest amount of insulation may be needed to limit boil-off to a rate below that needed to provide fuel vapor to the engines in cruise. On the ground, a system to capture boil-off liquified gas fuel may be provided. Another characteristic requirements for insulation includes surface area of a fuel tank 108. A very large tank may provide a large volume of liquified gas fuel per unit surface area. Boil-off rate may be determined by heat transferred into fuel tank 108. Heat transferred into the fuel tank 108 may be a function of tank surface area and/or insulation effectiveness. For example, heat transferred into fuel tank 108 may be proportional to tank surface area and/or inversely proportional to insulation effectiveness. For a given insulation and storage volume, fuel tank 108 having a larger surface area may result in more boil-off. Or, for a given boil-off rate and storage volume, fuel

tank 108 having a larger surface area will need more insulation. As explained above, degree of insulation may be selected to provide a desired rate of boil-off. Insulation may also include a chamber located between the inner wall and the outer wall foams, aerogels, reflective materials, and the like of fuel tank 108. Chamber may contain gas such as air, nitrogen, argon, or the like. In some cases, gas may be actively pumped into the chamber to ensure that the gas within the chamber is clean and dry and thereby not conducive to condensation, freezing or contamination.

[0038] Continuing to refer to FIG. 1, fuel tank 108 of the plurality of fuel tanks are located aft of the cabin in the main body. As used herein, “cabin” is the portion of the aircraft that holds the crew, passengers, and cargo. Fuel tanks may be stored at least partially aft of the cabin, near the propulsors. For the purposes of this disclosure, the fuel tanks are stored at least partially aft of the cabin if at least a portion of the fuel tanks extends behind or “aft” of the rearmost portion of cabin. In an embodiment, because of the low density of liquified gas fuel, storing fuel tanks behind the cabin of the aircraft 112 may not substantially affect the longitudinal center of gravity of the aircraft 112. FIG. 5 shows a depiction of a multi-lobe tank configuration. In a multi-lobe tank configuration, in some embodiments, each fuel tank 108 does not form a completely circular shape, such that the surface of fuel tank 108 may deform to achieve tension. Multi-lobe tank configuration may be beneficial as it provides more tank volume compared to a singular spherical tank. Multi-lobe tank configuration may be derived by adding tanks in the junctions between tanks. Junctions are discussed in further detail in FIG. 13. Fuel tanks, as discussed above, may provide structural support to the aircraft by acting as load bearing columns between the floor and ceiling of the aircraft 112. In an embodiment, fuel tanks may be mounted within the airframe. As used herein, an “airframe” provides structure to an aircraft. Airframe may be a part of the structural components of the aircraft 112. The plurality of fuel tanks may span across the full width of the aircraft 112. In embodiments, as a non-limiting example where the ceiling of the aircraft is downward sloping, each fuel tank 108 of the plurality of fuel tanks may vary in diameter and length. Fuel tanks may be stored vertically or horizontally. In an embodiment and as described above, fuel tank 108 may be a vertically oriented multi-lobe tank wherein its pressurized walls and its septa may extend from a lower outer mold line to an upper outer mold line of a fuel tank 108. A top and bottom of fuel tank 108 may be closed out by spherical end caps inset from the outer mold line to provide room for the outer mold line skin’s supporting structure. This may assist aircraft 112 in resisting pressurization and simultaneously carrying the shear stress that is otherwise carried by the ribs of the airframe of aircraft 112. Fuel tanks may be stored in rows aft of the cabin. In an embodiment, there may be two rows of fuel tanks. In doubling the rows, each fuel tank 108 had a smaller tank wall radius, therefore each fuel tank wall is thinner and lighter.

[0039] With continued reference to FIG. 1, a computing device 104 may engage a flight preparation 116 as a function of a fill datum or a desired level. As used in the current disclosure, a “flight preparation” is a process of preparing the fuel tank for flight. The flight preparation 116 may include detaching fuel line 120 and vent line 124 as a function of fill datum of desired level. Fuel line 120 and vent

line 124 may be attached when fill datum indicates fuel tank 108 while actively engaged in the fueling process. Fuel line 120 and vent line 124 may be detached when fill datum indicates fueling process has been completed or the plane is in flight or flight preparations. The fueling process is generally completed when the fuel tank 108 is full or has enough fuel to complete one or a plurality of flight plans. The flight preparation 116 may include adjusting the vent for flight conditions, so that the vent may safely vent fuel gasses during flight. The flight preparation 116 may include capping the fuel tank. In some cases, flight preparation 116 may include preparing actively conditioning a cavity of the fuel tank. For example, the fuel tank may include a chamber 142 between an inner wall and an outer wall of the tank. In some cases, clean dry gas (e.g., air, nitrogen, argon, etc.) may be pumped into the chamber 142. The clean dry gas may prevent condensation and freezing of humidity against the inner wall, thereby aiding in insulating the liquified gas fuel. The clean dry air may also be vented from the chamber 142 and/or recirculated.

[0040] Continuing to refer to FIG. 1, a fuel line 120 may be used to transfer liquid nitrogen fuel into fuel tank 108. As used in the current disclosure, a “fuel line” is a hose configured to transport fuel from a fuel reservoir to a fuel tank. In embodiments, a fuel line may be comprised of a hose, tube, or conduit. A fuel line may be made of rubber, plastic, steel, copper, or any combination thereof. Fuel lines may be configured to be able to contain pressurized liquified gas fuel. In some embodiments, liquified gas fuel may include liquid hydrogen fuel. Fuel line 120 may be in fluid connection with a liquified gas reservoir. Liquified gas fuel may be pumped from a liquified gas reservoir through fuel line 120 into fuel tank 108. The fuel line 120 may be connected to the fuel tank 108 through an airtight or a watertight seal. A seal may be made of a rubber compound. In embodiments, Fuel line 120 may be removably attached to a Fuel tank 108. For the purposes of this disclosure, “removably attached” means attached to an object such that it may be removed without damaging the object. The attachment of the fuel line 120 may be determined as a function fill datum. Computing device 104 may be detach a fuel line 120 unlocking the latching mechanism between fuel line 120 and fuel tank 108. The latching mechanism may only be unlocked when there is no fuel passing through the fuel line 120. Once the latching mechanism is unlocked the fuel line 120 may be configured to automatically be stored on a fuel line storage device. This device may store fuel line 120 on a reel or spool as a function of fill datum.

[0041] Continuing to refer to FIG. 1, a vent line 124 in fluid connection with a fuel tank 108. As used in this disclosure, elements are in “fluid connection” when a fluid may move from one element to another, notwithstanding any temporary obstructions such as valves. In embodiments, a vent line 124 may be configured to be attached to a vent of a fuel tank 108. A vent line may be configured to capture gas from fuel tank 108 as a result of a vent boil off. A vent line 124 may be made of rubber, plastic, steel, copper, or any combination thereof. Vent lines 124 may be configured to be able to contain pressurized liquified gas fuel. Vent line 124 may be in fluid connection with a reservoir of liquified gas. In embodiments, Vent line 124 may be removably attached to a fuel tank 108. The attachment of the fuel line 120 may be determined as a function fill datum.

[0042] Continuing to reference FIG. 1, fuel tank 108 is connected to a vent line 124. As used herein, a “vent line” is a tether or a bundle of tethers, e.g., hose, tubing, cables, wires, and the like, which is configured to removably attach with a mating component of a vent of a fuel tank 108. Vent line 124 may be made of a rigid or a flexible material. For example, vent line 124 may be composed of polypropylene, polycarbonate, acrylonitrile butadiene styrene, polyethylene, nylon, polystyrene, polyether ether ketone, and the like. Vent line 124 may also be composed of metals such as carbon fiber, aluminum, titanium, copper, or the like. Vent line 124 may be attached to a vent located at the top of the tank 108 such that vent line 124 directs gaseous fuel from the vent to an external fuel tank. External fuel tank may be consistent with any fuel tank as disclosed herein. Vent line 124 may be configured to direct gaseous fuel overboard, wherein overboard means off of the aircraft. While aircraft is grounded, aircraft engine and other systems are not using hydrogen vapor. Because of this, boil-off from the liquid fuel is collected through a vent line 124 to an external fuel tank.

[0043] Continuing to reference FIG. 1, vent line 124 may be insulated to prevent the accumulation of frozen gases, such as frozen water vapor. The boiling point of hydrogen is -423.2 F, which may be the freezing point of other gases. For example, water/water vapor freezes at 32 F. In an embodiment, insulation may include vermiculite, fiberglass, X-Aerogel, crosslinked Aerogel, firebrick, or any other insulation with a low thermal conductivity coefficient. A low thermal conductivity may be any thermal conductivity less than 0.10 , 0.01 , or 0.001 W/m-k. In this disclosure, “insulation” is a component or layer configured to reduce heat transfer. Insulation may limit heat transfer by conduction. Insulation may surround the vent line. In some embodiments, insulation may surround the circumference of vent line. In some embodiments, insulation may partially surround the circumference of the vent line. Alternatively or additionally, vent line 124 may include an inner wall and an outer wall, wherein a vacuum resides in between. As used herein, an “inner wall” is the inner barrier of a vent line that would be in contact with the gaseous fuel. As used herein, an “outer wall” is the outer barrier of a vent line that is in contact with an environment outside the vent line. In some embodiments, there may be insulation between the outer and inner wall. In some embodiments, there may be a vacuum as an insulation between the outer and inner wall.

[0044] With continued reference to FIG. 1, additionally or alternatively, a heat exchanger may be disposed proximal vent line 124. As used in this disclosure, a “heat exchanger” is a device that is configured to transfer heat from a first fluid or medium to a second fluid or medium. In some cases, a heat exchanger may include a hot fluid channel, which may be in thermal communication with the vent line 124. A “channel,” as used herein, is a component that is substantially impermeable to gases and contains and/or directs a flow of hot fluid. A Channel may act as a heat exchanger between the hot fluid and the cold gaseous fuel. Hot fluids may include oils, air, water, propylene glycol, ethylene glycol, or the like. Additionally or alternatively, heat may be provided by a powered heat source such as a radiator, electric resistance heater, combustion heater, or the like. Powered heat source may be placed in thermal communication with the vent line 124, for instance to heat the cold gaseous fuel. Powered heat source may be placed in thermal communication with hot fluid channel to heat the hot fluid

that may, for example, heat the cold gaseous fuel. In some embodiments, powered heat source may be in thermal communication with a length of vent line, for example to heat the cold gaseous fuel. In some embodiments, powered heat source may be in thermal communication with a length of hot fluid channel, for example, to heat the hot fluid. In some embodiments, heat elements may be in thermal communication with the vent line to prevent an accumulation of frozen air and/or water vapor.

[0045] With continued reference to FIG. 1, fuel tank 108 may be configured to be filled to a desired level 132. As used in the current disclosure, a “desired level” is when the tank is filled with enough fuel for a plurality of flights 136 plus additional reserve fuel. A sensor 128 may be used to determine when a tank 108 is at a desired level 132. A desired level 132 may include enough fuel to several flights in one day. In an embodiment, a desired level 132 may include filling the tank to capacity or filling the tank to a predetermined fraction of the capacity of the fuel tank 108. Desired level 132 may be determined as a function of weight or refueling time. Reserve fuel may include approximately 10%-20% more fuel than what is required to complete a plurality of flights. A desired level 132 may additionally comprise filling the fuel tank to capacity. As used in the current disclosure, “filling the fuel tank” comprises transferring fuel from a fuel reservoir into a fuel tank. Fill datum may also be considered when determining if a tank is at a desired level 132.

[0046] With continued reference to FIG. 1, method 100 includes a sensor 128. Sensor 128 may be connected to fuel line 120, vent line 124, fuel tank 108, or components thereof. As used in this disclosure, a “sensor” is a device that is configured to detect a physical characteristic and/or a phenomenon and convert the detection into a signal, such as an output sensor signal. In one or more embodiments, sensor 128 may be configured to transmit information, such as fill datum, related to a detection. As used in the current disclosure, a “fill datum” is a datum related to the fueling process of fuel tank 108. For example, and without limitation, fill datum may include information related to failure of the fuel tank 108, fuel line 120, and vent line 124. In some embodiments, fill datum may be related to any abnormal phenomena, such as and without limitation, temperature, voltage, current, pressure, fuel levels and the like as it relates to fuel tank 108. For instance, and without limitation, fill datum may include a high/low temperature of liquified gas fuel or fuel tank 108 directly. In another instance, and without limitation, sensor 128 may detect various characteristics of fuel line 120, vent line 124, fuel tank 108, and/or their surrounding environment that allows for fill datum to be indirectly determined, such as by computing device 104. For example, and without limitation, sensor 128 may determine a voltage, current, and thermal conditions of a surrounding environment of fuel tank 108 to determine temperature of liquified gas fuel. Fill datum may additionally include the amount of fuel currently in fuel tank 108 in relation to the fuel tank’s 108 capacity. Fuel datum may also be the rate or volume of fuel that is passing through fuel line 120. This may be done using a flow sensor placed within fuel line 120. Or alternatively, the amount of fuel to complete a flight plan or a plurality of flight plans. As another non-limiting example, fill datum may include a pressure of fuel tank 108 that that is generated as a function of liquid nitrogen fuel. Fill datum may include an indication that a fuel tank 108

may need to be prepared for flight. Fill datum may be related to the fuel tank 108 needing maintenance. Fill datum may also include the weight of the fuel. For example, and without limitation, sensor 128 may include a humidistat, hygrometer, voltage sensor, current sensor, multimeter, voltmeter, ammeter, electrical current sensor, resistance sensor, impedance sensor, capacitance sensor, a Wheatstone bridge, displacements sensor, vibration sensor, Daly detector, electroscope, electron multiplier, Faraday cup, galvanometer, Hall effect sensor, Hall probe, magnetic sensor, optical sensor, magnetometer, magnetoresistance sensor, MEMS magnetic field sensor, metal detector, planar Hall sensor, thermal sensor, thermocouple, resistance thermometer, semiconductor-based temperature sensors, thermistor, fuel flow sensor, fuel level sensor, and the like, among others. Sensor(s) 128 may efficaciously include, without limitation, any of the sensors disclosed in the entirety of the present disclosure.

[0047] With continued reference to FIG. 1, in one or more embodiments, sensor 128 may be communicatively connected to computing device 104, fuel line 120, vent line 124, and fuel tank 108. As used in this disclosure, “communicatively connected” means connected by way of a connection, attachment, or linkage between two or more relata which allows for reception and/or transmittance of information therebetween. For example, and without limitation, this connection may be wired or wireless, direct, or indirect, and between two or more components, circuits, devices, systems, and the like, which allows for reception and/or transmittance of data and/or signal(s) therebetween. Data and/or signals therebetween may include, without limitation, electrical, electromagnetic, magnetic, video, audio, radio, and microwave data and/or signals, combinations thereof, and the like, among others. A communicative connection may be achieved, for example and without limitation, through wired or wireless electronic, digital, or analog, communication, either directly or by way of one or more intervening devices or components. Further, communicative connection may include electrically coupling or connecting at least an output of one device, component, or circuit to at least an input of another device, component, or circuit. For example, and without limitation, via a bus or other facility for intercommunication between elements of a computing device. Communicative connecting may also include indirect connections via, for example and without limitation, wireless connection, radio communication, low power wide area network, optical communication, magnetic, capacitive, or optical coupling, and the like. In some instances, the terminology “communicatively coupled” may be used in place of communicatively connected in this disclosure.

[0048] With continued reference to FIG. 1, in one or more embodiments, sensor 128 may include one or more sensors. In some embodiments, sensor 128 may be a sensor suite. For example, and without limitation, sensor 128 may include a plurality of sensors, such as for redundancy purposes or confidence level reassurance of each detection by a sensor of the plurality of sensors. Sensor 128 may be a contact or a non-contact sensor. For example, and without limitation, sensor 128 may be physically attached to fuel line 120, vent line 124, and/or fuel tank 108. In other embodiments, sensor 128 may be remote to fuel line 120, vent line 124, and fuel tank 108.

[0049] With continued reference to FIG. 1, in one or more embodiments, sensor 128 may include a temperature sensor or probe. A temperature sensor may include a thermocouple,

thermometer, pyrometer, resistance temperature detector (RTD), platinum resistance temperature detector (PRTD), thermistor, negative temperature coefficient (NTC) thermistor, semiconductor based integrated circuit (IC), microbolometers, local temperature sensor, remote digital temperature sensor, infrared sensor, infrared or visible spectrum imaging device, thermophile infrared sensor, any combination thereof, and the like. Temperature, for the purposes of this disclosure, and as would be appreciated by someone of ordinary skill in the art, is a measure of the heat energy of a system. Temperature, as measured by any number or combinations of sensors present within sensor 128, may be measured in Fahrenheit (° F.), Celsius (° C.), Kelvin (° K.), or another scale alone or in combination. The temperature measured by sensors may comprise electrical signals, such as output sensor signal, which are transmitted to their appropriate destination, such as computing device 104, using a wireless and/or wired connection.

[0050] With continued reference to FIG. 1, in other embodiments, sensor 128 may also include other sensors to detect the amount of fuel in fuel tank 108. This may occur by measuring the amount of fuel that pass through a fuel line 120. This may also take into account the amount of fuel that has vented through vent line 124. In embodiments, Sensor 128 may include a hydrogen sensor. A hydrogen sensor may be configured to detect both gaseous and liquid hydrogen. Sensor 128 may be configured to determine the capacity of fuel tank 108 for liquid hydrogen. Sensor 128 may be able to determine the amount of liquid hydrogen in the fuel tank 108. In one or more embodiments, sensor 128 may include a plurality of independent sensors, where any number of the described sensors may be used to detect any number of physical quantities associated with liquid/gaseous hydrogen fuel, fuel line 120, vent line 124, fuel tank 108, and/or components thereof. Independent sensors may include separate sensors measuring physical quantities that may be powered by and/or in communication with circuits independently, where each may signal sensor output to a control circuit such as a user graphical interface. In an embodiment, use of a plurality of independent sensors may result in redundancy configured to employ more than one sensor that measures the same phenomenon, those sensors being of the same type, a combination of, or another type of sensor not disclosed, so that in the event one sensor fails, the ability of sensor 128 to detect phenomenon, such as fill datum of fuel tank 108, may be maintained.

[0051] With continued reference to FIG. 1, in some embodiments, sensor 128 may include a pressure sensor. “Pressure,” for the purposes of this disclosure, and as would be appreciated by someone of ordinary skill in the art, is a measure of force required to stop a fluid from expanding and is usually stated in terms of force per unit area. The pressure sensor that may be included in sensor 128 may be configured to measure an atmospheric pressure and/or a change of atmospheric pressure. In some embodiments, the pressure sensor may include an absolute pressure sensor, a gauge pressure sensor, a vacuum pressure sensor, a differential pressure sensor, a sealed pressure sensor, and/or other unknown pressure sensors or alone or in a combination thereof. The pressure sensor may include a barometer. In some embodiments, the pressure sensor may be used to indirectly measure fuel flow, Fuel tank level, liquified gas fuel level, and altitude. In some embodiments, the pressure sensor may be configured to transform a pressure into an

analogue electrical signal. In some embodiments, the pressure sensor may be configured to transform a pressure into a digital signal.

[0052] In one or more embodiments, sensor 128 may include a moisture sensor. “Moisture,” as used in this disclosure, is the presence of water, which may include vaporized water in air, condensation on the surfaces of objects, or concentrations of liquid water. Moisture may include humidity. “Humidity,” as used in this disclosure, is the property of a gaseous medium (almost always air) to hold water in the form of vapor.

[0053] With continued reference to FIG. 1, in one or more embodiments, sensor 128 may include a sensor suite which may include an array of sensors that may detect similar or unique phenomena. For example, in a non-limiting embodiment, sensor suite may include a plurality of temperature sensors, such as, for example, thermometers or a mixture of thermistors and thermometers. Method 100 may include a plurality of sensors in the form of individual sensors or a sensor suite working in tandem or individually. A sensor suite may include a plurality of independent sensors, as described in this disclosure, where any number of the described sensors may be used to detect any number of physical or electrical quantities associated with a fuel tank 108 of an aircraft. Independent sensors may include separate sensors measuring physical or electrical quantities that may be powered by and/or in communication with circuits independently, where each may signal sensor output to a control circuit such as computing device 104. In an embodiment, use of a plurality of independent sensors may result in redundancy configured to employ more than one sensor that measures the same phenomenon, those sensors being of the same type, a combination of, or another type of sensor not disclosed, so that in the event one sensor fails, the ability to detect a phenomenon is maintained.

[0054] With continued reference to FIG. 1, fuel tank 108 may be configured hold enough fuel to successfully complete a plurality of flights 136. As used in the current disclosure, a “plurality of flights” may include conducting two or more flights in succession using the aircraft. In some embodiments, aircraft may receive substantially no fuel in between flights of the plurality of flights. In embodiments, a fuel tank 108 may be configured to refuel prior to the start of the plurality of flights 136 (and after the end of the plurality of flights). In other embodiments, the fuel tank 108 may be configured to refuel then make a plurality of flights 136 without refueling in between flights. A benefit of this embodiment may include shorter layover times between the plurality of flights 136, because the aircraft does not need to refuel. Once the plurality of flights 136 has been completed the aircraft may then return to the original airport/refueling location for refueling or another airport that offers liquified gas fuel. The plurality of flights 136 may begin and end at the same location or airport. In an embodiment, the plurality of flights 136 may begin at a first refueling station and terminate at a second at a refueling station. As used in the current disclosure, a “refueling station” is an airport that is equipped to refuel aircraft that use liquified gas fuel. The tank 108 may only be filled at locations with a refueling station. In an embodiment, a first refueling station may be located in California and a second refueling station may be located in Texas. A multitude of flights may occur in between the aircrafts departure from a first refueling station to its arrival at a second refueling station. A first and second

refueling station may be located at the same or different airports. Refueling commence at any time after the arrival of the aircraft at the second refueling station. Refueling may occur at a continuous rate over an extended period. This may include refueling the fuel tank **108** overnight or refueling over several hours ranging from 5-10 hours. Refueling may also take place in 30 minutes or more. Once the tanks have been refueled, the aircraft then may be prepared for flight. In some embodiments, methods presently described provide an improvement of flight fueling regimes, as changeover times may be reduced and more flights operating on liquified gas fuel may be offered (even between airports which lack those refueling capabilities). In some cases, a computing device may be configured to determine desired level **132**, for example as a function of plurality of flights, fuel consumption, and fuel reserves.

[0055] Now referring to FIG. 2, shown is an exemplary embodiment of fuel tank **108**. A key component that may heavily impact shape of fuel tank **108** is pressure. Fuel tank **108** may be far lighter if pressure is resisted in tension as compared to bending. In tension, thin-walled tanks may be used. Thin-walled tanks may be lighter than thick-walled tanks as less material is used. In some embodiments, tension in a tank may be generally achieved by shapes that provide a circular cross section, including spheres, cylinders, and cones. In an embodiment, a pressurized tank of a given volume may be made as a sphere to place the tank in tension. Alternatively, a tank made as a cube may require tank walls to operate in bending; thus, the cube tank would likely be vastly heavier than a sphere tank of similar volume. Cube tanks may be heavier as thick walls may be necessary to resist bending. Accordingly, in some embodiments, any tank geometry may provide tank walls acting in tension. Tank geometry is further described in FIGS. 3-12.

[0056] Still referring to FIG. 2, the walls of fuel tank **108** may operate at or below a limit stress. A “limit stress” is a threshold stress below which fuel tank **108** can operate at to avoid failure or damage. Stress may be defined in hoop direction. Hoop stress occurs along the circumference of a tank. Stress of a thin-wall cylindrical tank may be calculated in the hoop direction by multiplying the pressure (e.g., in lb./in²) by the radius (e.g., in inches), and then dividing that value by the tank wall thickness (e.g., in inches). Also as shown below, the stress of a thin-wall cylindrical tank may be calculated in the longitudinal direction by dividing the hoop stress in half. The mathematical equation for stress may be found by first calculating a force by multiplying π by the radius (r) squared and the pressure (P). The force is then divided by $2 \cdot \pi \cdot r \cdot t$, the area of stress with a thickness (t) to calculate the stress. Thus, the equation simplifies to half the hoop stress, or $P \cdot r / 2 \cdot t$. The maximum stress in a cylindrical tank is the vector sum of the hoop and longitudinal stresses. A thin-wall hemisphere provides equal stress everywhere of $P \cdot r / 2 \cdot t$. A cylindrical tank may be fabricated with hemispherical end caps. If the end caps have twice the radius of the cylinder, and the skin thickness is everywhere the same, then the stresses in all parts of the tank may be similar. In an embodiment, the fuel tanks of the plurality of fuel tanks each include domed end caps. Domed end caps may have twice the radius of the cylindrical portion of the fuel tank **108**, therefore allowing the fuel tanks to have equal stress and pressure throughout the whole tank.

$$\text{Force} = \pi r^2 P$$

$$\text{Stress} = \frac{\pi r^2 P}{2 \pi r t} = \frac{Pr}{2t}$$

[0057] Now referring to FIGS. 3-13, exemplary tank geometries are illustrated. Fuel tank **108** may include a shape, or tank geometry, having a plurality of curved surfaces. All tank geometries may exist in tension, and fuel tank **108** may have any of the tank geometries described herein.

[0058] Now referring to just FIG. 3, an isometric view of an exemplary embodiment of a conical tank **300** is illustrated. Fuel tank **108** may be a conical tank. A “conical” tank is a type of tapered tank that has a cone-shaped tank geometry. A “tapered tank” as used herein, is a shape of a tank that has a larger cross section at one end of the tank then the other end, which has a smaller cross-section, and continuously transitions from the larger cross-section to the smaller cross-section. FIG. 3 shows a tapered tank wherein the left face of it is considered the “top of the tank” and the right side is considered the “bottom of the tank”. A tapered tank may facilitate an easy outflow of liquids from the bottom of the tank. Conical pure-tension tank shapes may include spherical and cylindrical shapes, possibly with spherical end caps. For example, a tank could be conical with a spherical end cap. This might resemble an ice cream cone. This cone may be truncated, with another spherical end cap on the opposing end. In an embodiment, two or more conical tanks may be combined to form a multi-lobe geometry. Multi-lobe geometry is discussed in further detail in FIG. 11-12.

[0059] Now referring to FIG. 4, an isometric view of an exemplary embodiment of a curved axisymmetric tank **400** is exhibited. Fuel tank **108** may be a curved axisymmetric tank **400**. Another type of tapered tank, a “curved axisymmetric tank” has a circular cross-section with compound curvature on the sides or ends. Curved axisymmetric tank may be capped with hemispheres. A curved axisymmetric tank may more efficiently fill a volume with variable depth.

[0060] Now referring to FIG. 5, an isometric view of an exemplary embodiment of a double-curved tank **500** is presented. Fuel tank **108** may be a double-curved tank. A “double-curved tank” occurs when a tapered tank is merged with a similar or mirror-image tank with a central septum. The two tanks comprising a double-curved tank may be intersected along their length and a septum may be placed at the tank junction to address the resulting tension. In this disclosure, a “central septum” is a partition centrally located in a system separating two compartments. In some cases, central septum may not be parallel to the tank axis; for example, it may be favorable to fill a volume of constant width with a curved, variable-height ceiling. Tank axis may then be adjusted so that the tank wall on the outer side of the tank may be a selected distance from the compartment wall.

[0061] Now referring to FIG. 6, a double-curved tank need not have identical compartments; a front quarter view of an exemplary embodiment of a dual tank with different diameters **600** is illustrated. Fuel tank **108** may be a dual tank with different diameters. In some compartments in this embodiment, height on one side of the compartment may be lower than on the other. Two or more compartments in fuel tank **108** may have different diameters if the ceiling height is different across the compartment. The two or more tanks

may be joined with one or more septa that may form a curved surface as seen in top view, see cambered tanks below. In an embodiment, the shapes of the compartments may differ. In this case, the two merged tanks (dual tank) may have different diameters to maximize their height along their length.

[0062] Now referring to FIG. 7, some tapered tanks may be sheared so that instead of following a straight centerline, fuel tank 108 follows a curved camber line. Shown in FIG. 7 is a quarter side view of an exemplary embodiment of a cambered, tapered tank 700. The camber line enables the tank to more efficiently fit a volume with variable depth but one flat side, for example a floor, while conforming more closely to, for example, a curved ceiling. This results in a centerline that may be curved as seen in the figure. An example may be shown in wireframe and surfaced views. This tank geometry may have a flat bottom and a curved top.

[0063] Now referring to FIGS. 8A and 8B, an isometric and side quarter views of an exemplary embodiment of a dual-cambered tank is presented, respectively. Fuel tank 108 may be a dual-cambered tank 800. A “dual-cambered” tank is the same as normal cambered tank, but the camber line may be curved from the top view as well as the side view. Two cambered lines can place the outer surface of the tank at a selected distance from the compartment wall. The bottom and right edge of fuel tank 108 may be straight while the top view and side view of camber line may be curved. Circular cross sections of the tank may be sheared so that they remain circles in the lateral-vertical plane, for example. Alternatively, circular cross sections may be orthogonal to the camber line. From a stress standpoint, wherever the camber line curvature of the tank is modest, the stress difference is probably very small.

[0064] Now referring to FIG. 9, a quarter front view of an exemplary embodiment of a double tank is shown. The double tank 900, as explained above, may have a curved septum separating the two compartments of fuel tank 108. A double tank may be a multi-lobe tank.

[0065] Now referring to FIGS. 10 and 11, a front quarter view of an exemplary embodiment of a multi-lobe tank is illustrated in both figures. Given a rectangular compartment cross-section with a longitudinally oriented tank, this cross-section may be occupied by a single circular cross-section tank. Or, as noted above, a double tank may be used to provide greater cross-section area within the rectangular compartment. Additional spheres may be added to fill in the four corners. A “multi-lobe tank” is a tank that has more than two compartments/curvatures attached together. In some cases, two more spheres may be added to fill in the valleys between the two main tanks. Multi-lobe tanks may have any number of compartments/curvatures, but there may be a diminishing return on increasing complexity; either engineering judgment or actual engineering may be applied. In an embodiment, a multi-lobe tank may have four lobes added to fill in the corners of a notional envelope indicated by the lines in the figure. A “lobe” as used herein, are curved sections of a multi-lobe geometry. This provides a more valuable tank volume for a given compartment volume. Each compartment may have a circular cross section as shown, which may be trimmed to the large, main lobes. The main lobes may then be trimmed to small lobes. Each junction may be then faced with a septum. This “trimming” can be seen in FIG. 3, wherein the multi-lobe tank 1100 is

trimmed with septa. On the other hand, the multi-lobe tank 1000 in FIG. 3 does not have trimmings. Fuel tank 108 may or may not have trimming.

[0066] Now referring to FIG. 12, an isometric view of an exemplary embodiment of a tank 1200 with a plurality of tank support links is shown. Fuel tank 108 needs to be mounted into aircraft 112, usually inside the body, in order to be a permanently attached. As explained above, fuel tank 108 may be a permanent tank, meaning it may be mounted to the aircraft for an extended period of time. Fuel tank 108 and aircraft 112 may be constructed separately, and then fuel tank 108 may be mounted within. In an embodiment, fuel tank 108 and BWB 1404 may be independent because the structural load paths of Method 100 may not pass through fuel tank 108, and the structural load paths of fuel tank 108 may not pass through BWB 1404, except insofar as necessary to restrain fuel tank 108 within BWB 1404. Fuel tank 108 may alternatively, or additionally, provide additional structural support to the airframe. Given this constraint, there may be at least two methods for tank support links to mount fuel tank 108 to BWB 1404: rigid and linked mounts.

[0067] Still referring to FIG. 12, fuel tank 108 may be mounted to BWB 1404 using a rigid mount. Fuel tank 108 may be a standalone structure and may be connected to the BWB 1404 with one or more rigid connections. A “rigid mount” or “rigid connection” is a type of link that does not allow for free movement in any direction; the link is rigid and/or non-movable. For example, if fuel tank 108 is a longitudinally-mounted cylindrical tank, it may have a series of feet on either side, which may be connected to the structure of fuel tank 108. These feet may also be connected, for example, to a compartment floor structure. Additionally, the rigid mount may impose loads on BWB 1404 and fuel tank 108. In an embodiment, during a flight maneuver, Method 100 may stretch, compress, or deform slightly, which may ultimately and slightly alter the location of the example feet mounting points. Overall, this ensues some deformation of the structure of fuel tank 108 and may impose additional loads on the tank that result in an unfavorably heavier design. Additional feet may be provided to distribute the load of fuel tank 108 more widely into BWB 1404. For example, feet may also connect the tank to compartment walls, the compartment ceiling structure, and the compartment aft pressure bulkhead. This arrangement may not be intended to reinforce BWB 1404 by its connection to fuel tank 108. Also, it may not be intended to reinforce fuel tank 108 by its connection to BWB 1404.

[0068] Still referring to FIG. 12, fuel tank 108 may be mounted to BWB 1404 using a linked mount. A standalone tank structure may be connected to the airframe in such a way that airframe deformation may not result in tank deformation, and vice-versa. This can be achieved through two ways to mount the tank: through its feet as described above, and through a series of links that may have hinges or ball-joints. A “linked mount” is a way to attach an object to another object that allows movement in one or more directions. With reference to the feet of fuel tank 108 as described above, to avoid structure deformation, the feet may be mounted to the airframe with rubber fittings that provide compliance. Exemplary feet include mounts manufactured by LORD Corp., of Williston, Vermont. A series of links that may have hinges or ball-joints may also mount fuel tank 108 to Method 100 without structural deformation occurring. For example, and as shown in the figure, fuel tank 108 may be

mounted at three points forming a triangle: a first point may be a rigid connection that provides location in three axes, a second point may be a link that provides substantially two (vertical and lateral) location, and a third point may be a link that provides substantially one (vertical) location. These three points are seen in the figure as links **1204**, **1208**, and **1212**, respectively. Link **1204** may support vertically, laterally, and longitudinally. Link **1208** may support vertically and laterally because it may be pivoted about a lateral axis. Link **1212** may provide only vertical support because it has ball joints at each end. A combination of links **1204** and **1208** may resist yaw and pitch motion. A combination of links **1208** and **1212** may resist roll. Altogether, motion of fuel tank **108** may be restrained against motion and rotation through the use of tank support links. A relative change in length between fuel tank **108** and BWB **1404** may be accommodated by Links **1208** and **1212** pivoting fore-aft. A relative change in width may be accommodated by Link **1212** pivoting laterally. A relative change in height may be unconstrained. Torsion applied to fuel tank **108** by Links **1208** and three may be accommodated by a spherical or cylindrical connection at link **1204**. One skilled in the art can provide alternate ways to achieve these objectives.

[0069] Now referring to FIG. **13**, an exemplary embodiment of a cross-sectional view of multi-lobe tanks that may be placed aft of the main body of aircraft **112**. The multi-lobe geometry of the multi-lobe tanks **1300** may provide tension for each fuel tank **108** of the plurality of multi-lobe tanks. Tension may be achieved by equal pressure in each lobe of the geometry. Multi-lobe geometry may include convex junctions joining each lobe. Convex junction **1316** can be seen in FIG. **13**.

[0070] Referring to FIG. **14**, an exemplary blended wing aircraft **1400** is illustrated. Aircraft **1400** may include a blended wing body **1404**. For the purposes of this disclosure, a “blended wing body aircraft” is an aircraft having a blended wing body. As used in this disclosure, A “blended wing body” (BWB), also known as a “blended body” or a “hybrid wing body” (HWB), is a fixed-wing aircraft body having no clear or abrupt demarcation between wings and a main body of the aircraft along a leading edge of the aircraft. For example, a BWB **1404** aircraft may have distinct wing and body structures, which are smoothly blended together with no clear dividing line or boundary feature between wing and fuselage. This contrasts with a flying wing, which has no distinct fuselage, and a lifting body, which has no distinct wings. A BWB **1404** design may or may not be tailless. One potential advantage of a BWB **1404** may be to reduce wetted area and any accompanying drag associated with a conventional wing-body junction. In some cases, a BWB **1404** may also have a wide airfoil-shaped body, allowing entire aircraft to generate lift and thereby facilitate reduction in size and/or drag of wings. In some cases, a BWB **1404** may be understood as a hybrid shape that resembles a flying wing, but also incorporates features from conventional aircraft. In some cases, this combination may offer several advantages over conventional tube-and-wing airframes. In some cases, a BWB airframe **1404** may help to increase fuel economy and create larger payload (cargo or passenger) volumes within the BWB. BWB **1404** may allow for advantageous interior designs. For instance, cargo can be loaded and/or passengers can board from the front or rear of the aircraft. A cargo or passenger area may be distributed across a relatively wide (when compared to conventional

tube-wing aircraft) fuselage, providing a large usable volume. In some embodiments, passengers seated within an interior of aircraft, real-time video at every seat can take place of window seats.

[0071] With continued reference to FIG. **14**, BWB **1404** of aircraft **1400** may include a nose portion. A “nose portion,” for the purposes of this disclosure, refers to any portion of aircraft **1400** forward of the aircraft’s fuselage. Nose portion may comprise a cockpit (for manned aircraft), canopy, aerodynamic fairings, windshield, and/or any structural elements required to support mechanical loads. Nose portion may also include pilot seats, control interfaces, gages, displays, inceptor sticks, throttle controls, collective pitch controls, and/or communication equipment, to name a few. Nose portion may comprise a swing nose configuration. A swing nose may be characterized by an ability of the nose to move, manually or automatically, into a differing orientation than its flight orientation to provide an opening for loading a payload into aircraft fuselage from the front of the aircraft. Nose portion may be configured to open in a plurality of orientations and directions.

[0072] With continued reference to FIG. **14**, BWB **1404** may include at least a structural component of aircraft **1400**. Structural components may provide physical stability during an entirety of an aircraft’s **1400** flight envelope, while on ground, and during normal operation. Structural components may comprise struts, beams, formers, stringers, longerons, interstitials, ribs, structural skin, doubles, straps, spars, or panels, to name a few. Structural components may also comprise pillars. In some cases, for the purpose of aircraft cockpits comprising windows/windshields, pillars may include vertical or near vertical supports around a window configured to provide extra stability around weak points in a vehicle’s structure, such as an opening where a window is installed. Where multiple pillars are disposed in an aircraft’s **1400** structure, they may be so named A, B, C, and so on named from nose to tail. Pillars, like any structural element, may be disposed a distance away from each other, along an exterior of aircraft **1400** and BWB **1404**. Depending on manufacturing method of BWB **1404**, pillars may be integral to frame and skin, comprised entirely of internal framing, or alternatively, may be only integral to structural skin elements. Structural skin will be discussed in greater detail below.

[0073] With continued reference to FIG. **14**, BWB **1404** may include a plurality of materials, alone or in combination, in its construction. At least a BWB **1404**, in an illustrative embodiment may include a welded steel tube frame further configured to form a general shape of a nose corresponding to an arrangement of steel tubes. Steel may include any of a plurality of alloyed metals, including but not limited to, a varying amount of manganese, nickel, copper, molybdenum, silicon, and/or aluminum, to name a few. Welded steel tubes may be covered in any of a plurality of materials suitable for aircraft skin. Some of these may include carbon fiber, fiberglass panels, cloth-like materials, aluminum sheeting, or the like. BWB **1404** may comprise aluminum tubing mechanically coupled in various and orientations. Mechanical fastening of aluminum members (whether pure aluminum or alloys) may comprise temporary or permanent mechanical fasteners appreciable by one of ordinary skill in the art including, but not limited to, screws, nuts and bolts, anchors, clips, welding, brazing, crimping, nails, blind rivets, pull-through rivets, pins, dowels, snap-

fits, clamps, and the like. BWB **1404** may additionally or alternatively use wood or another suitably strong yet light material for an internal structure.

[0074] With continued reference to FIG. **14**, aircraft **1400** may include monocoque or semi-monocoque construction. BWB **1404** may include carbon fiber. Carbon fiber may include carbon fiber reinforced polymer, carbon fiber reinforced plastic, or carbon fiber reinforced thermoplastic (e.g., CFRP, CRP, CFRTP, carbon composite, or just carbon, depending on industry). “Carbon fiber,” as used in this disclosure, is a composite material including a polymer reinforced with carbon. In general, carbon fiber composites consist of two parts, a matrix, and a reinforcement. In carbon fiber reinforced plastic, the carbon fiber constitutes the reinforcement, which provides strength. The matrix can include a polymer resin, such as epoxy, to bind reinforcements together. Such reinforcement achieves an increase in CFRP’s strength and rigidity, measured by stress and elastic modulus, respectively. In embodiments, carbon fibers themselves can each comprise a diameter between 5-10 micrometers and include a high percentage (i.e. above 85%) of carbon atoms. A person of ordinary skill in the art will appreciate that the advantages of carbon fibers include high stiffness, high tensile strength, low weight, high chemical resistance, high temperature tolerance, and low thermal expansion. According to embodiments, carbon fibers may be combined with other materials to form a composite, when permeated with plastic resin and baked, carbon fiber reinforced polymer becomes extremely rigid. Rigidity may be considered analogous to stiffness which may be measured using Young’s Modulus. Rigidity may be defined as a force necessary to bend and/or flex a material and/or structure to a given degree. For example, ceramics have high rigidity, which can be visualized by shattering before bending. In embodiments, carbon fibers may additionally, or alternatively, be composited with other materials like graphite to form reinforced carbon-carbon composites, which include high heat tolerances over 2000° C. A person of skill in the art will further appreciate that aerospace applications may require high-strength, low-weight, high heat resistance materials in a plurality of roles, such as without limitation fuselages, fairings, control surfaces, and structures, among others.

[0075] With continued reference to FIG. **14**, BWB **1404** may include at least a fuselage. A “fuselage,” for the purposes of this disclosure, refers to a main body of an aircraft **1400**, or in other words, an entirety of the aircraft **1400** except for nose, wings, empennage, nacelles, and control surfaces. In some cases, fuselage may contain an aircraft’s payload. At least a fuselage may comprise structural components that physically support a shape and structure of an aircraft **1400**. Structural components may take a plurality of forms, alone or in combination with other types. Structural components vary depending on construction type of aircraft **1400** and specifically, fuselage. A fuselage **1412** may include a truss structure. A truss structure may be used with a lightweight aircraft. A truss structure may include welded steel tube trusses. A “truss,” as used in this disclosure, is an assembly of beams that create a rigid structure, for example without limitation including combinations of triangles to create three-dimensional shapes. A truss structure may include wood construction in place of steel tubes, or a combination thereof. In some embodiments, structural components can comprise steel tubes and/or wood beams. An

aircraft skin may be layered over a body shape constructed by trusses. Aircraft skin may comprise a plurality of materials such as plywood sheets, aluminum, fiberglass, and/or carbon fiber.

[0076] With continued reference to FIG. **14**, in embodiments, at least a fuselage may comprise geodesic construction. Geodesic structural elements may include stringers wound about formers (which may be alternatively called station frames) in opposing spiral directions. A “stringer,” for the purposes of this disclosure is a general structural element that includes a long, thin, and rigid strip of metal or wood that is mechanically coupled to and spans the distance from, station frame to station frame to create an internal skeleton on which to mechanically couple aircraft skin. A former (or station frame) can include a rigid structural element that is disposed along a length of an interior of a fuselage orthogonal to a longitudinal (nose to tail) axis of aircraft **1400**. In some cases, a former forms a general shape of at least a fuselage. A former may include differing cross-sectional shapes at differing locations along a fuselage, as the former is a structural component that informs an overall shape of the fuselage. In embodiments, aircraft skin can be anchored to formers and strings such that an outer mold line of volume encapsulated by the formers and stringers comprises a same shape as aircraft **1400** when installed. In other words, former(s) may form a fuselage’s ribs, and stringers may form interstitials between the ribs. A spiral orientation of stringers about formers may provide uniform robustness at any point on an aircraft fuselage such that if a portion sustains damage, another portion may remain largely unaffected. Aircraft skin may be mechanically coupled to underlying stringers and formers and may interact with a fluid, such as air, to generate lift and perform maneuvers.

[0077] With continued reference to FIG. **14**, according to some embodiments, a fuselage can comprise monocoque construction. Monocoque construction can include a primary structure that forms a shell (or skin in an aircraft’s case) and supports physical loads. Monocoque fuselages are fuselages in which the aircraft skin or shell may also include a primary structure. In monocoque construction aircraft skin would support tensile and compressive loads within itself and true monocoque aircraft can be further characterized by an absence of internal structural elements. Aircraft skin in this construction method may be rigid and can sustain its shape with substantially no structural assistance from underlying skeleton-like elements. Monocoque fuselage may include aircraft skin made from plywood layered in varying grain directions, epoxy-impregnated fiberglass, carbon fiber, or any combination thereof.

[0078] With continued reference to FIG. **14**, according to some embodiments, a fuselage may include a semi-monocoque construction. Semi-monocoque construction, as used in this disclosure, is used interchangeably with partially monocoque construction, discussed above. In semi-monocoque construction, a fuselage may derive some structural support from stressed aircraft skin and some structural support from underlying frame structure made of structural components. Formers or station frames can be seen running transverse to a long axis of fuselage with circular cutouts which may be used in real-world manufacturing for weight savings and for routing of electrical harnesses and other modern on-board systems. In a semi-monocoque construction, stringers may be thin, long strips of material that run

parallel to a fuselage's long axis. Stringers can be mechanically coupled to formers permanently, such as with rivets. Aircraft skin can be mechanically coupled to stringers and formers permanently, such as by rivets as well. A person of ordinary skill in the art will appreciate that there are numerous methods for mechanical fastening of the aforementioned components like screws, nails, dowels, pins, anchors, adhesives like glue or epoxy, or bolts and nuts, to name a few. According to some embodiments, a subset of semi-monocoque construction may be unibody construction. Unibody, which is short for "unitized body" or alternatively "unitary construction," vehicles are characterized by a construction in which body, floor plan, and chassis form a single structure, for example an automobile. In the aircraft world, a unibody may include internal structural elements, like formers and stringers, constructed in one piece, integral to an aircraft skin. In some cases, stringers and formers may account for the bulk of any aircraft structure (excluding monocoque construction). Stringers and formers can be arranged in a plurality of orientations depending on aircraft operation and materials. Stringers may be arranged to carry axial (tensile or compressive), shear, bending or torsion forces throughout their overall structure. Due to their coupling to aircraft skin, aerodynamic forces exerted on aircraft skin may be transferred to stringers. Location of said stringers greatly informs type of forces and loads applied to each and every stringer, all of which may be accounted for through design processes including, material selection, cross-sectional area, and mechanical coupling methods of each member. Similar methods may be performed for former assessment and design. In general, formers may be significantly larger in cross-sectional area and thickness, depending on location, than stringers. Both stringers and formers may comprise aluminum, aluminum alloys, graphite epoxy composite, steel alloys, titanium, or an undisclosed material alone or in combination.

[0079] With continued reference to FIG. 14, in some cases, a primary purpose for a substructure of a semi-monocoque structure is to stabilize a skin. Typically, aircraft structure is required to have a very light weight and as a result, in some cases, aircraft skin may be very thin. In some cases, unless supported, this thin skin structure may tend to buckle and/or cripple under compressive and/or shear loads. In some cases, underlying structure may be primarily configured to stabilize skins. For example, in an exemplary conventional airliner, wing structure is an airfoil-shaped box with truncated nose and aft triangle; without stabilizing substructure, in some cases, this box would buckle upper skin of the wing and the upper skin would also collapse into the lower skin under bending loads. In some cases, deformations are prevented with ribs that support stringers which stabilize the skin. Fuselages are similar with bulkheads or frames, and stringers.

[0080] With continued reference to FIG. 14, in some embodiments, another common structural form is sandwich structure. As used in this disclosure, "sandwich structure" includes a skin structure having an inner and outer skin separated and stabilized by a core material. In some cases, sandwich structure may additionally include some number of ribs or frames. In some cases, sandwich structure may include metal, polymer, and/or composite. In some cases, core material may include honeycomb, foam plastic, and/or end-grain balsa wood. In some cases, sandwich structure can be popular on composite light airplanes, such as gliders and

powered light planes. In some cases, sandwich structure may not use stringers, and sandwich structure may allow number of ribs or frames to be reduced, for instance in comparison with a semi-monocoque structure. In some cases, sandwich structure may be suitable for smaller, possibly unmanned, unpressurized blended wing body aircraft.

[0081] With continued reference to FIG. 14, stressed skin, when used in semi-monocoque construction, may bear partial, yet significant, load. In other words, an internal structure, whether it be a frame of welded tubes, formers and stringers, or some combination, is not sufficiently strong enough by design to bear all loads. The concept of stressed skin is applied in monocoque and semi-monocoque construction methods of at least a fuselage and/or BWB **1404**. In some cases, monocoque may be considered to include substantially only structural skin, and in that sense, aircraft skin undergoes stress by applied aerodynamic fluids imparted by fluid. Stress as used in continuum mechanics can be described in pound-force per square inch (lbf/in²) or Pascals (Pa). In semi-monocoque construction stressed skin bears part of aerodynamic loads and additionally imparts force on an underlying structure of stringers and formers.

[0082] With continued reference to FIG. 14, a fuselage may include an interior cavity. An interior cavity may include a volumetric space configurable to house passenger seats and/or cargo. An interior cavity may be configured to include receptacles for fuel tanks, batteries, fuel cells, or other energy sources as described herein. In some cases, a post may be supporting a floor (i.e., deck), or in other words a surface on which a passenger, operator, passenger, payload, or other object would rest on due to gravity when within an aircraft **1400** is in its level flight orientation or sitting on ground. A post may act similarly to stringer in that it is configured to support axial loads in compression due to a load being applied parallel to its axis due to, for example, a heavy object being placed on a floor of aircraft **1400**. A beam may be disposed in or on any portion a fuselage that requires additional bracing, specifically when disposed transverse to another structural element, like a post, which would benefit from support in that direction, opposing applied force. A beam may be disposed in a plurality of locations and orientations within a fuselage as necessitated by operational and constructional requirements.

[0083] With continued reference to FIG. 14, aircraft **1400** may include at least a flight component **1408**. A flight component **1408** may be consistent with any description of a flight component described in this disclosure, such as without limitation propulsors, control surfaces, rotors, paddle wheels, engines, propellers, wings, winglets, or the like. For the purposes of this disclosure, at least a "flight component" is at least one element of an aircraft **1400** configured to manipulate a fluid medium such as air to propel, control, or maneuver an aircraft. In nonlimiting examples, at least a flight component may include a rotor mechanically connected to a rotor shaft of an electric motor further mechanically affixed to at least a portion of aircraft **1400**. In some embodiments, at least a flight component **1408** may include a propulsor, for example a rotor attached to an electric motor configured to produce shaft torque and in turn, create thrust. As used in this disclosure, an "electric motor" is an electrical machine that converts electric energy into mechanical work.

[0084] With continued reference to FIG. 14, for the purposes of this disclosure, "torque", is a twisting force that

tends to cause rotation. Torque may be considered an effort and a rotational analogue to linear force. A magnitude of torque of a rigid body may depend on three quantities: a force applied, a lever arm vector connecting a point about which the torque is being measured to a point of force application, and an angle between the force and the lever arm vector. A force applied perpendicularly to a lever multiplied by its distance from the lever's fulcrum (the length of the lever arm) is its torque. A force of three newtons applied two meters from the fulcrum, for example, exerts the same torque as a force of one newton applied six meters from the fulcrum. In some cases, direction of a torque can be determined by using a right-hand grip rule which states: if fingers of right hand are curled from a direction of lever arm to direction of force, then thumb points in a direction of the torque. One of ordinary skill in the art would appreciate that torque may be represented as a vector, consistent with this disclosure, and therefore may include a magnitude and a direction. "Torque" and "moment" are used interchangeably within this disclosure. Any torque command or signal within this disclosure may include at least the steady state torque to achieve the torque output to at least a propulsor.

[0085] With continued reference to FIG. 14, at least a flight component may be one or more devices configured to affect aircraft's **1400** attitude. "Attitude," for the purposes of this disclosure, is the relative orientation of a body, in this case aircraft **1400**, as compared to earth's surface or any other reference point and/or coordinate system. In some cases, attitude may be displayed to pilots, personnel, remote users, or one or more computing devices in an attitude indicator, such as without limitation a visual representation of a horizon and its relative orientation to aircraft **1400**. A plurality of attitude datums may indicate one or more measurements relative to an aircraft's pitch, roll, yaw, or throttle compared to a relative starting point. One or more sensors may measure or detect an aircraft's **1400** attitude and establish one or more attitude datums. An "attitude datum," for the purposes of this disclosure, refers to at least an element of data identifying an attitude of an aircraft **1400**.

[0086] With continued reference to FIG. 14, in some cases, aircraft **1400** may include one or more of an angle of attack sensor and a yaw sensor. In some embodiments, one or more of an angle of attack sensor and a yaw sensor may include a vane (e.g., wind vane). In some cases, vane may include a protrusion on a pivot with an aft tail. The protrusion may be configured to rotate about pivot to maintain zero tail angle of attack. In some cases, pivot may turn an electronic device that reports one or more of angle of attack and/or yaw, depending on, for example, orientation of the pivot and tail. Alternatively or additionally, in some cases, one or more of angle of attack sensor and/or yaw sensor may include a plurality of pressure ports located in selected locations, with pressure sensors located at each pressure port. In some cases, differential pressure between pressure ports can be used to estimate angle of attack and/or yaw.

[0087] With continued reference to FIG. 14, in some cases, aircraft **1400** may include at least a pilot control. As used in this disclosure, a "pilot control," is an interface device that allows an operator, human or machine, to control a flight component of an aircraft. Pilot control may be communicatively connected to any other component presented in aircraft **1400**, the communicative connection may include redundant connections configured to safeguard

against single-point failure. In some cases, a plurality of attitude datums may indicate a pilot's instruction to change heading and/or trim of an aircraft **1400**. Pilot input may indicate a pilot's instruction to change an aircraft's pitch, roll, yaw, throttle, and/or any combination thereof. Aircraft trajectory may be manipulated by one or more control surfaces and propulsors working alone or in tandem consistent with the entirety of this disclosure. "Pitch," for the purposes of this disclosure refers to an aircraft's angle of attack, which is a difference between a plane including at least a portion of both wings of the aircraft running nose to tail and a horizontal flight trajectory. For example, an aircraft may pitch "up" when its nose is angled upward compared to horizontal flight, as in a climb maneuver. In another example, an aircraft may pitch "down," when its nose is angled downward compared to horizontal flight, like in a dive maneuver. In some cases, angle of attack may not be used as an input, for instance pilot input, to any system disclosed herein; in such circumstances proxies may be used such as pilot controls, remote controls, or sensor levels, such as true airspeed sensors, pitot tubes, pneumatic/hydraulic sensors, and the like. "Roll" for the purposes of this disclosure, refers to an aircraft's position about its longitudinal axis, which is to say that when an aircraft rotates about its axis from its tail to its nose, and one side rolls upward, as in a banking maneuver. "Yaw," for the purposes of this disclosure, refers to an aircraft's turn angle, when an aircraft rotates about an imaginary vertical axis intersecting center of earth and aircraft **1400**. "Throttle," for the purposes of this disclosure, refers to an aircraft outputting an amount of thrust from a propulsor. In context of a pilot input, throttle may refer to a pilot's input to increase or decrease thrust produced by at least a propulsor. Flight components **1408** may receive and/or transmit signals, for example an aircraft command signal. Aircraft command signal may include any signal described in this disclosure, such as without limitation electrical signal, optical signal, pneumatic signal, hydraulic signal, and/or mechanical signal. In some cases, an aircraft command may be a function of a signal from a pilot control. In some cases, an aircraft command may include or be determined as a function of a pilot command. For example, aircraft commands may be determined as a function of a mechanical movement of a throttle. Signals may include analog signals, digital signals, periodic or aperiodic signal, step signals, unit impulse signal, unit ramp signal, unit parabolic signal, signum function, exponential signal, rectangular signal, triangular signal, sinusoidal signal, sinc function, or pulse width modulated signal. Pilot control may include circuitry, computing devices, electronic components, or a combination thereof that translates pilot input into a signal configured to be transmitted to another electronic component. In some cases, a plurality of attitude commands may be determined as a function of an input to a pilot control. A plurality of attitude commands may include a total attitude command datum, such as a combination of attitude adjustments represented by one or a certain number of combinatorial datums. A plurality of attitude commands may include individual attitude datums representing total or relative change in attitude measurements relative to pitch, roll, yaw, and throttle.

[0088] With continued reference to FIG. 14, in some embodiments, pilot control may include at least a sensor. As used in this disclosure, a "sensor" is a device that detects a phenomenon. In some cases, a sensor may detect a phenom-

enon and transmit a signal that is representative of the phenomenon. At least a sensor may include, torque sensor, gyroscope, accelerometer, magnetometer, inertial measurement unit (IMU), pressure sensor, force sensor, proximity sensor, displacement sensor, vibration sensor, among others. At least a sensor may include a sensor suite which may include a plurality of sensors that may detect similar or unique phenomena. For example, in a non-limiting embodiment, sensor suite may include a plurality of accelerometers, a mixture of accelerometers and gyroscopes, or a mixture of an accelerometer, gyroscope, and torque sensor. For the purposes of the disclosure, a “torque datum” is one or more elements of data representing one or more parameters detailing power output by one or more propulsors, flight components, or other elements of an aircraft. A torque datum may indicate the torque output of at least a flight component **1408**. At least a flight component **1408** may include any propulsor as described herein. In embodiment, at least a flight component **1408** may include an electric motor, a propeller, a jet engine, a paddle wheel, a rotor, turbine, or any other mechanism configured to manipulate a fluid medium to propel an aircraft as described herein, an embodiment of at least a sensor may include or be included in, a sensor suite. The herein disclosed system and method may comprise a plurality of sensors in the form of individual sensors or a sensor suite working in tandem or individually. A sensor suite may include a plurality of independent sensors, as described herein, where any number of the described sensors may be used to detect any number of physical or electrical quantities associated with an aircraft power system or an electrical energy storage system. Independent sensors may include separate sensors measuring physical or electrical quantities that may be powered by and/or in communication with circuits independently, where each may signal sensor output to a control circuit such as a user graphical interface. In a non-limiting example, there may be four independent sensors housed in and/or on battery pack measuring temperature, electrical characteristic such as voltage, amperage, resistance, or impedance, or any other parameters and/or quantities as described in this disclosure. In an embodiment, use of a plurality of independent sensors may result in redundancy configured to employ more than one sensor that measures the same phenomenon, those sensors being of the same type, a combination of, or another type of sensor not disclosed, so that in the event one sensor fails, the ability of a battery management system and/or user to detect phenomenon is maintained and in a non-limiting example, a user alter aircraft usage pursuant to sensor readings.

[0089] With continued reference to FIG. **14**, at least a sensor may include a moisture sensor. “Moisture,” as used in this disclosure, is the presence of water, this may include vaporized water in air, condensation on the surfaces of objects, or concentrations of liquid water. Moisture may include humidity. “Humidity,” as used in this disclosure, is the property of a gaseous medium (almost always air) to hold water in the form of vapor. An amount of water vapor contained within a parcel of air can vary significantly. Water vapor is generally invisible to the human eye and may be damaging to electrical components. There are three primary measurements of humidity, absolute, relative, specific humidity. “Absolute humidity,” for the purposes of this disclosure, describes the water content of air and is expressed in either grams per cubic meters or grams per

kilogram. “Relative humidity,” for the purposes of this disclosure, is expressed as a percentage, indicating a present stat of absolute humidity relative to a maximum humidity given the same temperature. “Specific humidity,” for the purposes of this disclosure, is the ratio of water vapor mass to total moist air parcel mass, where parcel is a given portion of a gaseous medium. A moisture sensor may be psychrometer. A moisture sensor may be a hygrometer. A moisture sensor may be configured to act as or include a humidistat. A “humidistat,” for the purposes of this disclosure, is a humidity-triggered switch, often used to control another electronic device. A moisture sensor may use capacitance to measure relative humidity and include in itself, or as an external component, include a device to convert relative humidity measurements to absolute humidity measurements. “Capacitance,” for the purposes of this disclosure, is the ability of a system to store an electric charge, in this case the system is a parcel of air which may be near, adjacent to, or above a battery cell.

[0090] With continued reference to FIG. **14**, at least a sensor may include electrical sensors. An electrical sensor may be configured to measure voltage across a component, electrical current through a component, and resistance of a component. Electrical sensors may include separate sensors to measure each of the previously disclosed electrical characteristics such as voltmeter, ammeter, and ohmmeter, respectively. One or more sensors may be communicatively coupled to at least a pilot control, the manipulation of which, may constitute at least an aircraft command. Signals may include electrical, electromagnetic, visual, audio, radio waves, or another undisclosed signal type alone or in combination. At least a sensor communicatively connected to at least a pilot control may include a sensor disposed on, near, around or within at least pilot control. At least a sensor may include a motion sensor. “Motion sensor,” for the purposes of this disclosure refers to a device or component configured to detect physical movement of an object or grouping of objects. One of ordinary skill in the art would appreciate, after reviewing the entirety of this disclosure, that motion may include a plurality of types including but not limited to: spinning, rotating, oscillating, gyrating, jumping, sliding, reciprocating, or the like. At least a sensor may include, torque sensor, gyroscope, accelerometer, torque sensor, magnetometer, inertial measurement unit (IMU), pressure sensor, force sensor, proximity sensor, displacement sensor, vibration sensor, among others. At least a sensor may include a sensor suite which may include a plurality of sensors that may detect similar or unique phenomena. For example, in a non-limiting embodiment, sensor suite may include a plurality of accelerometers, a mixture of accelerometers and gyroscopes, or a mixture of an accelerometer, gyroscope, and torque sensor. The herein disclosed system and method may comprise a plurality of sensors in the form of individual sensors or a sensor suite working in tandem or individually. A sensor suite may include a plurality of independent sensors, as described herein, where any number of the described sensors may be used to detect any number of physical or electrical quantities associated with an aircraft power system or an electrical energy storage system. Independent sensors may include separate sensors measuring physical or electrical quantities that may be powered by and/or in communication with circuits independently, where each may signal sensor output to a control circuit such as a user graphical interface. In an embodiment, use of a plurality

of independent sensors may result in redundancy configured to employ more than one sensor that measures the same phenomenon, those sensors being of the same type, a combination of, or another type of sensor not disclosed, so that in the event one sensor fails, the ability to detect phenomenon is maintained and in a non-limiting example, a user alter aircraft usage pursuant to sensor readings.

[0091] With continued reference to FIG. 14, at least a flight component **1408** may include wings, empennages, nacelles, control surfaces, fuselages, and landing gear, among others, to name a few. In embodiments, an empennage may be disposed at the aftmost point of an aircraft body **1404**. Empennage may comprise a tail of aircraft **1400**, further comprising rudders, vertical stabilizers, horizontal stabilizers, stabilators, elevators, trim tabs, among others. At least a portion of empennage may be manipulated directly or indirectly by pilot commands to impart control forces on a fluid in which the aircraft **1400** is flying. Manipulation of these empennage control surfaces may, in part, change an aircraft's heading in pitch, roll, and yaw. Wings comprise may include structures which include airfoils configured to create a pressure differential resulting in lift. Wings are generally disposed on a left and right side of aircraft **1400** symmetrically, at a point between nose and empennage. Wings may comprise a plurality of geometries in planform view, swept wing, tapered, variable wing, triangular, oblong, elliptical, square, among others. Wings may be blended into the body of the aircraft such as in a BWB **1404** aircraft **1400** where no strong delineation of body and wing exists. A wing's cross section geometry may comprise an airfoil. An "airfoil" as used in this disclosure, is a shape specifically designed such that a fluid flowing on opposing sides of it exert differing levels of pressure against the airfoil. In embodiments, a bottom surface of an aircraft can be configured to generate a greater pressure than does a top surface, resulting in lift. A wing may comprise differing and/or similar cross-sectional geometries over its cord length, e.g. length from wing tip to where wing meets the aircraft's body. One or more wings may be symmetrical about an aircraft's longitudinal plane, which comprises a longitudinal or roll axis reaching down a center of the aircraft through the nose and empennage, and the aircraft's yaw axis. In some cases, wings may comprise controls surfaces configured to be commanded by a pilot and/or autopilot to change a wing's geometry and therefore its interaction with a fluid medium. Flight component **1408** may include control surfaces. Control surfaces may include without limitation flaps, ailerons, tabs, spoilers, and slats, among others. In some cases, control surfaces may be disposed on wings in a plurality of locations and arrangements. In some cases, control surfaces may be disposed at leading and/or trailing edges of wings, and may be configured to deflect up, down, forward, aft, or any combination thereof.

[0092] In some cases, flight component **1408** may include a winglet. For the purposes of this disclosure, a "winglet" is a flight component configured to manipulate a fluid medium and is mechanically attached to a wing or aircraft and may alternatively called a "wingtip device." Wingtip devices may be used to improve efficiency of fixed-wing aircraft by reducing drag. Although there are several types of wingtip devices which function in different manners, their intended effect may be to reduce an aircraft's drag by partial recovery of tip vortex energy. Wingtip devices can also improve

aircraft handling characteristics and enhance safety for aircraft **1400**. Such devices increase an effective aspect ratio of a wing without greatly increasing wingspan. Extending wingspan may lower lift-induced drag but would increase parasitic drag and would require boosting the strength and weight of the wing. As a result according to some aeronautic design equations, a maximum wingspan made be determined above which no net benefit exists from further increased span. There may also be operational considerations that limit the allowable wingspan (e.g., available width at airport gates).

[0093] Wingtip devices, in some cases, may increase lift generated at wingtip (by smoothing airflow across an upper wing near the wingtip) and reduce lift-induced drag caused by wingtip vortices, thereby improving a lift-to-drag ratio. This increases fuel efficiency in powered aircraft and increases cross-country speed in gliders, in both cases increasing range. U.S. Air Force studies indicate that a given improvement in fuel efficiency correlates directly and causally with increase in an aircraft's lift-to-drag ratio. The term "winglet" has previously been used to describe an additional lifting surface on an aircraft, like a short section between wheels on fixed undercarriage. An upward angle (i.e., cant) of a winglet, its inward or outward angle (i.e., toe), as well as its size and shape are selectable design parameters which may be chosen for correct performance in a given application. A wingtip vortex, which rotates around from below a wing, strikes a cambered surface of a winglet, generating a force that angles inward and slightly forward. A winglet's relation to a wingtip vortex may be considered analogous to sailboat sails when sailing to windward (i.e., close-hauled). Similar to the close-hauled sailboat's sails, winglets may convert some of what would otherwise-be wasted energy in a wingtip vortex to an apparent thrust. This small contribution can be worthwhile over the aircraft's lifetime. Another potential benefit of winglets is that they may reduce an intensity of wake vortices. Wake vortices may trail behind an aircraft **1400** and pose a hazard to other aircraft. Minimum spacing requirements between aircraft at airports are largely dictated by hazards, like those from wake vortices. Aircraft are classified by weight (e.g., "Light," "Heavy," and the like) often base upon vortex strength, which grows with an aircraft's lift coefficient. Thus, associated turbulence is greatest at low speed and high weight, which may be produced at high angle of attack near airports. Winglets and wingtip fences may also increase efficiency by reducing vortex interference with laminar airflow near wingtips, by moving a confluence of low-pressure air (over wing) and high-pressure air (under wing) away from a surface of the wing. Wingtip vortices create turbulence, which may originate at a leading edge of a wingtip and propagate backwards and inboard. This turbulence may delaminate airflow over a small triangular section of an outboard wing, thereby frustrating lift in that area. A fence/winglet drives an area where a vortex forms upward away from a wing surface, as the resulting vortex is repositioned to a top tip of the winglet.

[0094] With continued reference to FIG. 14, aircraft **1400** may include an energy source. Energy source may include any device providing energy to at least a flight component **1408**, for example at least a propulsors. Energy source may include, without limitation, a generator, a photovoltaic device, a fuel cell such as a hydrogen fuel cell, direct methanol fuel cell, and/or solid oxide fuel cell, or an electric energy storage device; electric energy storage device may

include without limitation a battery, a capacitor, and/or inductor. The energy source and/or energy storage device may include at least a battery, battery cell, and/or a plurality of battery cells connected in series, in parallel, or in a combination of series and parallel connections such as series connections into modules that are connected in parallel with other like modules. Battery and/or battery cell may include, without limitation, Li ion batteries which may include NCA, NMC, Lithium iron phosphate (LiFePO₄) and Lithium Manganese Oxide (LMO) batteries, which may be mixed with another cathode chemistry to provide more specific power if the application requires Li metal batteries, which have a lithium metal anode that provides high power on demand, Li ion batteries that have a silicon or titanite anode. In embodiments, the energy source may be used to provide electrical power to an electric or hybrid propulsor during moments requiring high rates of power output, including without limitation takeoff, landing, thermal de-icing, and situations requiring greater power output for reasons of stability, such as high turbulence situations. In some cases, battery may include, without limitation a battery using nickel based chemistries such as nickel cadmium or nickel metal hydride, a battery using lithium ion battery chemistries such as a nickel cobalt aluminum (NCA), nickel manganese cobalt (NMC), lithium iron phosphate (LiFePO₄), lithium cobalt oxide (LCO), and/or lithium manganese oxide (LMO), a battery using lithium polymer technology, lead-based batteries such as without limitation lead acid batteries, metal-air batteries, or any other suitable battery. A person of ordinary skill in the art, upon reviewing the entirety of this disclosure, will be aware of various devices of components that may be used as an energy source.

[0095] With continued reference to FIG. 14, in further nonlimiting embodiments, an energy source may include a fuel store. As used in this disclosure, a “fuel store” is an aircraft component configured to store a fuel. In some cases, a fuel store may include a fuel tank. Fuel may include a liquid fuel, a gaseous fluid, a solid fuel, and fluid fuel, a plasma fuel, and the like. As used in this disclosure, a “fuel” may include any substance that stores energy. Exemplary non-limiting fuels include hydrocarbon fuels, petroleum-based fuels, synthetic fuels, chemical fuels, Jet fuels (e.g., Jet-A fuel, Jet-B fuel, and the like), kerosene-based fuel, gasoline-based fuel, an electrochemical-based fuel (e.g., lithium-ion battery), a hydrogen-based fuel, natural gas-based fuel, and the like. As described in greater detail below fuel store may be located substantially within blended wing body 1404 of aircraft 1400, for example without limitation within a wing portion of blended wing body 1404. Aviation fuels may include petroleum-based fuels, or petroleum and synthetic fuel blends, used to power aircraft 1400. In some cases, aviation fuels may have more stringent requirements than fuels used for ground use, such as heating and road transport. Aviation fuels may contain additives to enhance or maintain properties important to fuel performance or handling. Fuel may be kerosene-based (JP-8 and Jet A-1), for example for gas turbine-powered aircraft. Piston-engine aircraft may use gasoline-based fuels and/or kerosene-based fuels (for example for Diesel engines). In some cases, specific energy may be considered an important criterion in selecting fuel for an aircraft 1400. Liquid fuel may include Jet-A. Presently Jet-A powers modern commercial airliners and is a mix of extremely refined kerosene and burns at temperatures at or above 49° C. (120° F.). Kerosene-based

fuel has a much higher flash point than gasoline-based fuel, meaning that it requires significantly higher temperature to ignite.

[0096] With continued reference to FIG. 14, modular aircraft 1400 may include an energy source which may include a fuel cell. As used in this disclosure, a “fuel cell” is an electrochemical device that combines a fuel and an oxidizing agent to create electricity. In some cases, fuel cells are different from most batteries in requiring a continuous source of fuel and oxygen (usually from air) to sustain the chemical reaction, whereas in a battery the chemical energy comes from metals and their ions or oxides that are commonly already present in the battery, except in flow batteries. Fuel cells can produce electricity continuously for as long as fuel and oxygen are supplied.

[0097] With continued reference to FIG. 14, in some embodiments, fuel cells may consist of different types. Commonly a fuel cell consists of an anode, a cathode, and an electrolyte that allows ions, often positively charged hydrogen ions (protons), to move between two sides of the fuel cell. At anode, a catalyst causes fuel to undergo oxidation reactions that generate ions (often positively charged hydrogen ions) and electrons. Ions move from anode to cathode through electrolyte. Concurrently, electrons may flow from anode to cathode through an external circuit, producing direct current electricity. At cathode, another catalyst causes ions, electrons, and oxygen to react, forming water and possibly other products. Fuel cells may be classified by type of electrolyte used and by difference in startup time ranging from 14 second for proton-exchange membrane fuel cells (PEM fuel cells, or PEMFC) to 10 minutes for solid oxide fuel cells (SOFC). In some cases, energy source may include a related technology, such as flow batteries. Within a flow battery fuel can be regenerated by recharging. Individual fuel cells produce relatively small electrical potentials, about 0.7 volts. Therefore, in some cases, fuel cells may be “stacked”, or placed in series, to create sufficient voltage to meet an application’s requirements. In addition to electricity, fuel cells may produce water, heat and, depending on the fuel source, very small amounts of nitrogen dioxide and other emissions. Energy efficiency of a fuel cell is generally between 40 and 90%.

[0098] Fuel cell may include an electrolyte. In some cases, electrolyte may define a type of fuel cell. Electrolyte may include any number of substances like potassium hydroxide, salt carbonates, and phosphoric acid. Commonly a fuel cell is fueled by hydrogen. Fuel cell may feature an anode catalyst, like fine platinum powder, which breaks down fuel into electrons and ions. Fuel cell may feature a cathode catalyst, often nickel, which converts ions into waste chemicals, with water being the most common type of waste. A fuel cell may include gas diffusion layers that are designed to resist oxidation.

[0099] With continued reference to FIG. 14, aircraft 1400 may include an energy source which may include a cell such as a battery cell, or a plurality of battery cells making a battery module. An energy source may be a plurality of energy sources. The module may include batteries connected in parallel or in series or a plurality of modules connected either in series or in parallel designed to deliver both the power and energy requirements of the application. Connecting batteries in series may increase the voltage of an energy source which may provide more power on demand. High voltage batteries may require cell matching when high peak

load is needed. As more cells are connected in strings, there may exist the possibility of one cell failing which may increase resistance in the module and reduce the overall power output as the voltage of the module may decrease as a result of that failing cell. Connecting batteries in parallel may increase total current capacity by decreasing total resistance, and it also may increase overall amp-hour capacity. The overall energy and power outputs of an energy source may be based on the individual battery cell performance, or an extrapolation based on the measurement of at least an electrical parameter. In an embodiment where an energy source includes a plurality of battery cells, the overall power output capacity may be dependent on the electrical parameters of each individual cell. If one cell experiences high self-discharge during demand, power drawn from an energy source may be decreased to avoid damage to the weakest cell. An energy source may further include, without limitation, wiring, conduit, housing, cooling system and battery management system. Persons skilled in the art will be aware, after reviewing the entirety of this disclosure, of many different components of an energy source.

[0100] With continued reference to FIG. 14, aircraft 1400 may include multiple flight component 1408 sub-systems, each of which may have a separate energy source. For instance, and without limitation, one or more flight components 1408 may have a dedicated energy source. Alternatively, or additionally, a plurality of energy sources may each provide power to two or more flight components 1408, such as, without limitation, a “fore” energy source providing power to flight components located toward a front of an aircraft 1400, while an “aft” energy source provides power to flight components located toward a rear of the aircraft 1400. As a further non-limiting example, a flight component of group of flight components may be powered by a plurality of energy sources. For example, and without limitation, two or more energy sources may power one or more flight components; two energy sources may include, without limitation, at least a first energy source having high specific energy density and at least a second energy source having high specific power density, which may be selectively deployed as required for higher-power and lower-power needs. Alternatively, or additionally, a plurality of energy sources may be placed in parallel to provide power to the same single propulsor or plurality of propulsors. Alternatively, or additionally, two or more separate propulsion subsystems may be joined using intertie switches (not shown) causing the two or more separate propulsion subsystems to be treatable as a single propulsion subsystem or system, for which potential under load of combined energy sources may be used as the electric potential. Persons skilled in the art, upon reviewing the entirety of this disclosure, will be aware of various combinations of energy sources that may each provide power to single or multiple propulsors in various configurations.

[0101] With continued reference to FIG. 14, aircraft 1400 may include a flight component 1408 that includes at least a nacelle. For the purposes of this disclosure, a “nacelle” is a streamlined body housing, which is sized according to that which is houses, such as without limitation an engine, a fuel store, or a flight component. When attached by a pylon entirely outside an airframe 1404 a nacelle may sometimes be referred to as a pod, in which case an engine within the nacelle may be referred to as a podded engine. In some cases an aircraft cockpit may also be housed in a nacelle, rather

than in a conventional fuselage. At least a nacelle may substantially encapsulate a propulsor, which may include a motor or an engine. At least a nacelle may be mechanically connected to at least a portion of aircraft 1400 partially or wholly enveloped by an outer mold line of the aircraft 1400. At least a nacelle may be designed to be streamlined. At least a nacelle may be asymmetrical about a plane comprising the longitudinal axis of the engine and the yaw axis of modular aircraft 1400.

[0102] With continued reference to FIG. 14, a flight component may include a propulsor. A “propulsor,” as used herein, is a component or device used to propel a craft by exerting force on a fluid medium, which may include a gaseous medium such as air or a liquid medium such as water. For the purposes of this disclosure, “substantially encapsulate” is the state of a first body (e.g., housing) surrounding all or most of a second body. A motor may include without limitation, any electric motor, where an electric motor is a device that converts electrical energy into mechanical work for instance by causing a shaft to rotate. A motor may be driven by direct current (DC) electric power; for instance, a motor may include a brushed DC motor or the like. A motor may be driven by electric power having varied or reversing voltage levels, such as alternating current (AC) power as produced by an alternating current generator and/or inverter, or otherwise varying power, such as produced by a switching power source. A motor may include, without limitation, a brushless DC electric motor, a permanent magnet synchronous motor, a switched reluctance motor, and/or an induction motor; persons skilled in the art, upon reviewing the entirety of this disclosure, will be aware of various alternative or additional forms and/or configurations that a motor may take or exemplify as consistent with this disclosure. In addition to inverter and/or switching power source, a circuit driving motor may include electronic speed controllers or other components for regulating motor speed, rotation direction, torque, and/or dynamic braking. Motor may include or be connected to one or more sensors detecting one or more conditions of motor; one or more conditions may include, without limitation, voltage levels, electromotive force, current levels, temperature, current speed of rotation, position sensors, and the like. For instance, and without limitation, one or more sensors may be used to detect back-EMF, or to detect parameters used to determine back-EMF, as described in further detail below. One or more sensors may include a plurality of current sensors, voltage sensors, and speed or position feedback sensors. One or more sensors may communicate a current status of motor to a flight controller and/or a computing device; computing device may include any computing device as described in this disclosure, including without limitation, a flight controller.

[0103] With continued reference to FIG. 14, a motor may be connected to a thrust element. Thrust element may include any device or component that converts mechanical work, for example of a motor or engine, into thrust in a fluid medium. Thrust element may include, without limitation, a device using moving or rotating foils, including without limitation one or more rotors, an airscrew or propeller, a set of airscrews or propellers such as contra-rotating propellers or co-rotating propellers, a moving or flapping wing, or the like. Thrust element may include without limitation a marine propeller or screw, an impeller, a turbine, a pump-jet, a paddle or paddle-based device, or the like. Thrust element

may include a rotor. Persons skilled in the art, upon reviewing the entirety of this disclosure, will be aware of various devices that may be used as thrust element. A thrust element may include any device or component that converts mechanical energy (i.e., work) of a motor, for instance in form of rotational motion of a shaft, into thrust within a fluid medium. As another non-limiting example, a thrust element may include an eight-bladed pusher propeller, such as an eight-bladed propeller mounted behind the engine to ensure the drive shaft is in compression.

[0104] With continued reference to FIG. 14, in nonlimiting embodiments, at least a flight component 1408 may include an airbreathing engine such as a jet engine, turbojet engine, turboshaft engine, ramjet engine, scramjet engine, hybrid propulsion system, turbofan engine, or the like. At least a flight component 1408 may be fueled by any fuel described in this disclosure, for instance without limitation Jet-A, Jet-B, diesel fuel, gasoline, or the like. In nonlimiting embodiments, a jet engine is a type of reaction engine discharging a fast-moving jet that generates thrust by jet propulsion. While this broad definition can include rocket, water jet, and hybrid propulsion, the term jet engine, in some cases, refers to an internal combustion airbreathing jet engine such as a turbojet, turbofan, ramjet, or pulse jet. In general, jet engines are internal combustion engines. As used in this disclosure, a “combustion engine” is a mechanical device that is configured to convert mechanical work from heat produced by combustion of a fuel. In some cases, a combustion engine may operate according to an approximation of a thermodynamic cycle, such as without limitation a Carnot cycle, a Cheng cycle, a Combined cycle, a Brayton cycle, an Otto cycle, an Allam power cycle, a Kalina cycle, a Rankine cycle, and/or the like. In some cases, a combustion engine may include an internal combustion engine. An internal combustion engine may include heat engine in which combustion of fuel occurs with an oxidizer (usually air) in a combustion chamber that comprises a part of a working fluid flow circuit. Exemplary internal combustion engines may without limitation a reciprocating engine (e.g., 4-stroke engine), a combustion turbine engine (e.g., jet engines, gas turbines, Brayton cycle engines, and the like), a rotary engine (e.g., Wankel engines), and the like. In nonlimiting embodiments, airbreathing jet engines feature a rotating air compressor powered by a turbine, with leftover power providing thrust through a propelling nozzle—this process may be known as a Brayton thermodynamic cycle. Jet aircraft may use such engines for long-distance travel. Early jet aircraft used turbojet engines that were relatively inefficient for subsonic flight. Most modern subsonic jet aircraft use more complex high-bypass turbofan engines. In some cases, they give higher speed and greater fuel efficiency than piston and propeller aeroengines over long distances. A few air-breathing engines made for highspeed applications (ramjets and scramjets) may use a ram effect of aircraft’s speed instead of a mechanical compressor. An airbreathing jet engine (or ducted jet engine) may emit a jet of hot exhaust gases formed from air that is forced into the engine by several stages of centrifugal, axial or ram compression, which is then heated and expanded through a nozzle. In some cases, a majority of mass flow through an airbreathing jet engine may be provided by air taken from outside of the engine and heated internally, using energy stored in the form of fuel. In some cases, a jet engine may include are turbofans. Alternatively and/or additionally, jet

engine may include a turbojets. In some cases, a turbofan may use a gas turbine engine core with high overall pressure ratio (e.g., 40:1) and high turbine entry temperature (e.g., about 1800 K) and provide thrust with a turbine-powered fan stage. In some cases, thrust may also be at least partially provided by way of pure exhaust thrust (as in a turbojet engine). In some cases, a turbofan may have a high efficiency, relative to a turbojet. In some cases, a jet engine may use simple ram effect (e.g., ramjet) or pulse combustion (e.g., pulsejet) to give compression. Persons skilled in the art, upon reviewing the entirety of this disclosure, will be aware of various devices that may be used as a thrust element.

[0105] With continued reference to FIG. 14, an aircraft 1400 may include a flight controller. As used in this disclosure, a “flight controller” is a device that generates signals for controlling at least a flight component 1408 of an aircraft 1400. In some cases, a flight controller includes electronic circuitry, such as without limitation a digital signal processor (DSP), an application-specific integrated circuit (ASIC), and/or a computing device. Flight controller may use sensor feedback to calculate performance parameters of motor, including without limitation a torque versus speed operation envelope. Persons skilled in the art, upon reviewing the entirety of this disclosure, will be aware of various devices and/or components that may be used as or included in a motor or a circuit operating a motor, as used, and described in this disclosure.

[0106] With continued reference to FIG. 14, computing device may include any computing device as described in this disclosure, including without limitation a microcontroller, microprocessor, digital signal processor (DSP) and/or system on a chip (SoC) as described in this disclosure. Computing device may include, be included in, and/or communicate with a mobile device such as a mobile telephone or smartphone. Computing device may include a single computing device operating independently, or may include two or more computing device operating in concert, in parallel, sequentially or the like; two or more computing devices may be included together in a single computing device or in two or more computing devices. Computing device may interface or communicate with one or more additional devices as described below in further detail via a network interface device. Network interface device may be utilized for connecting computing device to one or more of a variety of networks, and one or more devices. Examples of a network interface device include, but are not limited to, a network interface card (e.g., a mobile network interface card, a LAN card), a modem, and any combination thereof. Examples of a network include, but are not limited to, a wide area network (e.g., the Internet, an enterprise network), a local area network (e.g., a network associated with an office, a building, a campus or other relatively small geographic space), a telephone network, a data network associated with a telephone/voice provider (e.g., a mobile communications provider data and/or voice network), a direct connection between two computing devices, and any combinations thereof. A network may employ a wired and/or a wireless mode of communication. In general, any network topology may be used. Information (e.g., data, software etc.) may be communicated to and/or from a computer and/or a computing device. Computing device may include but is not limited to, for example, a computing device or cluster of computing devices in a first location and a second computing device or

cluster of computing devices in a second location. Computing device may include one or more computing devices dedicated to data storage, security, distribution of traffic for load balancing, and the like. Computing device may distribute one or more computing tasks as described below across a plurality of computing devices of computing device, which may operate in parallel, in series, redundantly, or in any other manner used for distribution of tasks or memory between computing devices. Computing device may be implemented using a “shared nothing” architecture in which data is cached at the worker, in an embodiment, this may enable scalability of system and/or computing device.

[0107] With continued reference to FIG. 14, computing device may be designed and/or configured to perform any method, method step, or sequence of method steps in any embodiment described in this disclosure, in any order and with any degree of repetition. For instance, computing device may be configured to perform a single step or sequence repeatedly until a desired or commanded outcome is achieved; repetition of a step or a sequence of steps may be performed iteratively and/or recursively using outputs of previous repetitions as inputs to subsequent repetitions, aggregating inputs and/or outputs of repetitions to produce an aggregate result, reduction or decrement of one or more variables such as global variables, and/or division of a larger processing task into a set of iteratively addressed smaller processing tasks. Computing device may perform any step or sequence of steps as described in this disclosure in parallel, such as simultaneously and/or substantially simultaneously performing a step two or more times using two or more parallel threads, processor cores, or the like; division of tasks between parallel threads and/or processes may be performed according to any protocol suitable for division of tasks between iterations. Persons skilled in the art, upon reviewing the entirety of this disclosure, will be aware of various ways in which steps, sequences of steps, processing tasks, and/or data may be subdivided, shared, or otherwise dealt with using iteration, recursion, and/or parallel processing.

[0108] Now referring to FIG. 15, a flow diagram for a method 1500 of method of use for fueling an aircraft. At step 1505, method 1500 includes storing liquified gas fuel using a fuel tank. Fuel tank may include any fuel tank, as described above in reference to FIGS. 1-14. Liquified gas fuel may include any liquified gas fuel, as described above in reference to FIGS. 1-14.

[0109] At step 1510, method 1500 includes fueling an aircraft using a fuel line. Fuel line may include any fuel line, as described above in reference to FIGS. 1-14. Fueling an aircraft may also include filling fuel tank with liquified gas fuel to a desired level, wherein a desired level comprises fuel for a plurality of flights plus reserves and removing the fuel line as function of the fuel level in the aircraft. Filling the fuel tank may occur as described above in reference to FIGS. 1-14.

[0110] At step 1515 method 1500 includes venting the fuel tank, using a vent line in fluid connection to the fuel tank, as described above in reference to FIGS. 1-14. Vent line may include any vent line, as described above in reference to FIGS. 1-14.

[0111] At step 1520 method 1500 includes preparing the fuel tank for flight as a function of a desired level, as described above in reference to FIGS. 1-14.

[0112] At step 1525 method 1500 includes flying, using the aircraft, a plurality of flights using the liquid gas fuel, as described above in reference to FIGS. 1-14. Plurality of flights may include any plurality of flights, as described above in reference to FIGS. 1-14. Plurality of flights may include flights to and/or from airports different from that which fuels the aircraft. In some cases, flights may include commercial airline flights. In some cases, plurality of flights may originate and terminate at same airport, where aircraft is fueled.

[0113] With continued reference to method 1500, the aircraft may include a blended wing body aircraft. The fuel tank may be configured to have a multi-lobe geometry. The fuel tank may include a septum. Additionally, the multi-lobe geometry may provide tension for each container of the multi-lobe geometry. The fill data may include the amount of fuel necessary for the flight. A flight preparation may be engaged as a function of a flight status. A vent line may be configured to capture gaseous hydrogen from the fuel tank. The vent line may be configured to release pressure from the fuel tank. The aircraft may be configured to engage in flight using liquified gas fuel.

[0114] It is to be noted that any one or more of the aspects and embodiments described herein may be conveniently implemented using one or more machines (e.g., one or more computing devices that are utilized as a user computing device for an electronic document, one or more server devices, such as a document server, etc.) programmed according to the teachings of the present specification, as will be apparent to those of ordinary skill in the computer art. Appropriate software coding can readily be prepared by skilled programmers based on the teachings of the present disclosure, as will be apparent to those of ordinary skill in the software art. Aspects and implementations discussed above employing software and/or software modules may also include appropriate hardware for assisting in the implementation of the machine executable instructions of the software and/or software module.

[0115] Such software may be a computer program product that employs a machine-readable storage medium. A machine-readable storage medium may be any medium that is capable of storing and/or encoding a sequence of instructions for execution by a machine (e.g., a computing device) and that causes the machine to perform any one of the methodologies and/or embodiments described herein. Examples of a machine-readable storage medium include, but are not limited to, a magnetic disk, an optical disc (e.g., CD, CD-R, DVD, DVD-R, etc.), a magneto-optical disk, a read-only memory “ROM” device, a random-access memory “RAM” device, a magnetic card, an optical card, a solid-state memory device, an EPROM, an EEPROM, and any combinations thereof. A machine-readable medium, as used herein, is intended to include a single medium as well as a collection of physically separate media, such as, for example, a collection of compact discs or one or more hard disk drives in combination with a computer memory. As used herein, a machine-readable storage medium does not include transitory forms of signal transmission.

[0116] Such software may also include information (e.g., data) carried as a data signal on a data carrier, such as a carrier wave. For example, machine-executable information may be included as a data-carrying signal embodied in a data carrier in which the signal encodes a sequence of instruction, or portion thereof, for execution by a machine (e.g., a

computing device) and any related information (e.g., data structures and data) that causes the machine to perform any one of the methodologies and/or embodiments described herein.

[0117] Examples of a computing device include, but are not limited to, an electronic book reading device, a computer workstation, a terminal computer, a server computer, a handheld device (e.g., a tablet computer, a smartphone, etc.), a web appliance, a network router, a network switch, a network bridge, any machine capable of executing a sequence of instructions that specify an action to be taken by that machine, and any combinations thereof. In one example, a computing device may include and/or be included in a kiosk.

[0118] FIG. 16 shows a diagrammatic representation of one embodiment of a computing device in the exemplary form of a computer system 1600 within which a set of instructions for causing a control system to perform any one or more of the aspects and/or methodologies of the present disclosure may be executed. It is also contemplated that multiple computing devices may be utilized to implement a specially configured set of instructions for causing one or more of the devices to perform any one or more of the aspects and/or methodologies of the present disclosure. Computer system 1600 includes a processor 1604 and a memory 1608 that communicate with each other, and with other components, via a bus 1612. Bus 1612 may include any of several types of bus structures including, but not limited to, a memory bus, a memory controller, a peripheral bus, a local bus, and any combinations thereof, using any of a variety of bus architectures.

[0119] Processor 1604 may include any suitable processor, such as without limitation a processor incorporating logical circuitry for performing arithmetic and logical operations, such as an arithmetic and logic unit (ALU), which may be regulated with a state machine and directed by operational inputs from memory and/or sensors; processor 1604 may be organized according to Von Neumann and/or Harvard architecture as a non-limiting example. Processor 1604 may include, incorporate, and/or be incorporated in, without limitation, a microcontroller, microprocessor, digital signal processor (DSP), Field Programmable Gate Array (FPGA), Complex Programmable Logic Device (CPLD), Graphical Processing Unit (GPU), general purpose GPU, Tensor Processing Unit (TPU), analog or mixed signal processor, Trusted Platform Module (TPM), a floating-point unit (FPU), and/or system on a chip (SoC).

[0120] Memory 1608 may include various components (e.g., machine-readable media) including, but not limited to, a random-access memory component, a read only component, and any combinations thereof. In one example, a basic input/output system 1616 (BIOS), including basic routines that help to transfer information between elements within computer system 1600, such as during start-up, may be stored in memory 1608. Memory 1608 may also include (e.g., stored on one or more machine-readable media) instructions (e.g., software) 1620 embodying any one or more of the aspects and/or methodologies of the present disclosure. In another example, memory 1608 may further include any number of program modules including, but not limited to, an operating system, one or more application programs, other program modules, program data, and any combinations thereof.

[0121] Computer system 1600 may also include a storage device 1624. Examples of a storage device (e.g., storage device 1624) include, but are not limited to, a hard disk drive, a magnetic disk drive, an optical disc drive in combination with an optical medium, a solid-state memory device, and any combinations thereof. Storage device 1624 may be connected to bus 1612 by an appropriate interface (not shown). Example interfaces include, but are not limited to, SCSI, advanced technology attachment (ATA), serial ATA, universal serial bus (USB), IEEE 1394 (FIREWIRE), and any combinations thereof. In one example, storage device 1624 (or one or more components thereof) may be removably interfaced with computer system 1600 (e.g., via an external port connector (not shown)). Particularly, storage device 1624 and an associated machine-readable medium 1628 may provide nonvolatile and/or volatile storage of machine-readable instructions, data structures, program modules, and/or other data for computer system 1600. In one example, software 1620 may reside, completely or partially, within machine-readable medium 1628. In another example, software 1620 may reside, completely or partially, within processor 1604.

[0122] Computer system 1600 may also include an input device 1632. In one example, a user of computer system 1600 may enter commands and/or other information into computer system 1600 via input device 1632. Examples of an input device 1632 include, but are not limited to, an alpha-numeric input device (e.g., a keyboard), a pointing device, a joystick, a gamepad, an audio input device (e.g., a microphone, a voice response system, etc.), a cursor control device (e.g., a mouse), a touchpad, an optical scanner, a video capture device (e.g., a still camera, a video camera), a touchscreen, and any combinations thereof. Input device 1632 may be interfaced to bus 1612 via any of a variety of interfaces (not shown) including, but not limited to, a serial interface, a parallel interface, a game port, a USB interface, a FIREWIRE interface, a direct interface to bus 1612, and any combinations thereof. Input device 1632 may include a touch screen interface that may be a part of or separate from display 1636, discussed further below. Input device 1632 may be utilized as a user selection device for selecting one or more graphical representations in a graphical interface as described above.

[0123] A user may also input commands and/or other information to computer system 1600 via storage device 1624 (e.g., a removable disk drive, a flash drive, etc.) and/or network interface device 1640. A network interface device, such as network interface device 1640, may be utilized for connecting computer system 1600 to one or more of a variety of networks, such as network 1644, and one or more remote devices 1648 connected thereto. Examples of a network interface device include, but are not limited to, a network interface card (e.g., a mobile network interface card, a LAN card), a modem, and any combination thereof. Examples of a network include, but are not limited to, a wide area network (e.g., the Internet, an enterprise network), a local area network (e.g., a network associated with an office, a building, a campus or other relatively small geographic space), a telephone network, a data network associated with a telephone/voice provider (e.g., a mobile communications provider data and/or voice network), a direct connection between two computing devices, and any combinations thereof. A network, such as network 1644, may employ a wired and/or a wireless mode of communication. In general,

any network topology may be used. Information (e.g., data, software **1620**, etc.) may be communicated to and/or from computer system **1600** via network interface device **1640**.

[0124] Computer system **1600** may further include a video display adapter **1652** for communicating a displayable image to a display device, such as display device **1636**. Examples of a display device include, but are not limited to, a liquid crystal display (LCD), a cathode ray tube (CRT), a plasma display, a light emitting diode (LED) display, and any combinations thereof. Display adapter **1652** and display device **1636** may be utilized in combination with processor **1604** to provide graphical representations of aspects of the present disclosure. In addition to a display device, computer system **1600** may include one or more other peripheral output devices including, but not limited to, an audio speaker, a printer, and any combinations thereof. Such peripheral output devices may be connected to bus **1612** via a peripheral interface **1656**. Examples of a peripheral interface include, but are not limited to, a serial port, a USB connection, a FIREWIRE connection, a parallel connection, and any combinations thereof.

[0125] The foregoing has been a detailed description of illustrative embodiments of the invention. Various modifications and additions can be made without departing from the spirit and scope of this invention. Features of each of the various embodiments described above may be combined with features of other described embodiments as appropriate in order to provide a multiplicity of feature combinations in associated new embodiments. Furthermore, while the foregoing describes a number of separate embodiments, what has been described herein is merely illustrative of the application of the principles of the present invention. Additionally, although particular methods herein may be illustrated and/or described as being performed in a specific order, the ordering is highly variable within ordinary skill to achieve methods, systems, and software according to the present disclosure. Accordingly, this description is meant to be taken only by way of example, and not to otherwise limit the scope of this invention.

[0126] Exemplary embodiments have been disclosed above and illustrated in the accompanying drawings. It will be understood by those skilled in the art that various changes, omissions, and additions may be made to that which is specifically disclosed herein without departing from the spirit and scope of the present invention.

What is claimed is:

1. A method for fueling an aircraft, wherein the method comprises:

- conditioning a fuel tank of an aircraft, wherein the fuel tank comprises a chamber between an inner wall and an outer wall of the fuel tank and clean dry gas is pumped into the chamber to aid in insulating a liquid gas fuel;
- venting the liquid gas fuel from the chamber, wherein a direction of fuel flow is regulated by a liquid check valve to prevent back flow of the vented liquid gas fuel to the fuel tank;
- storing, using the fuel tank, the liquid gas fuel;
- fueling the aircraft using a fuel line, wherein fueling comprises filling the fuel tank with the liquid gas fuel to a desired level, wherein a sensor connected to the fuel tank detects a plurality of fill data; and
- preparing the fuel tank for flight as a function of the desired level and the plurality of fill data.

2. The method of claim 1, wherein the aircraft comprises a blended wind body (BWB) aircraft.

3. The method of claim 1, wherein the fuel tank comprises an insulation material with a thermal conductivity of less than 0.10 W/m-K.

4. The method of claim 1, wherein the chamber comprises at least one layer of reflective foil within an insulative material.

5. The method of claim 1, wherein the fuel tank comprises at least a septum, wherein the septum divides the fuel tank into at least two compartments.

6. The method of claim 1, wherein the fuel tank comprises a plurality of fuel tanks divided by septa, wherein each fuel tank is divided by at least a septum.

7. The method of claim 1, wherein the fuel tank comprises a multi-lobe geometry, wherein the multi-lobe geometry provides tension for each lobe of the multi-lobe geometry.

8. The method of claim 1, wherein the fuel tank is mounted to the aircraft using a linked mount.

9. The method of claim 1, further comprising detecting a rate and volume of fuel passing through the fuel line using a flow sensor connected to the fuel line, wherein, the rate and volume of fuel passing through the fuel line contributes to the plurality of fill data.

10. The method of claim 1, further comprising detecting water vapor levels using the sensor connected to the fuel tank, wherein the sensor connected to the fuel tank comprises a moisture sensor, contributing to the plurality of fill data.

11. The method of claim 1, further comprising detecting an atmospheric pressure using the sensor connected to the fuel tank, wherein the sensor from the fuel tank comprises a pressure sensor, contributing to the plurality of fill data.

12. The method of claim 1, wherein the liquid gas fuel comprises liquid hydrogen.

13. The method of claim 1, wherein preparing the fuel tank as a function of the desired level and the plurality of fill data comprises detaching the fuel line, wherein detaching the fuel line comprises unlocking a latching mechanism between the fuel line and the fuel tank.

14. The method of claim 1, wherein:

the fuel tank further comprises a vent, wherein the vent is in fluid connection with the fuel tank and an interface port; and

venting the liquid gas fuel from the chamber further comprises venting fuel gases offboard the aircraft through the vent.

15. The method of claim 1, wherein:

the fuel tank further comprises a vent, wherein the vent is in fluid connection with the fuel tank and an interface port; and

venting the liquid gas fuel from the chamber further comprises:

connecting a vent line to the vent at the interface port; and

collecting fuel gases in an external fuel tank as a function of venting the liquid gas fuel from the chamber.

16. The method of claim 15, further comprising regulating a temperature of the liquid gas fuel using a heat exchanger, wherein the heat exchanger:

comprises a fluid channel in thermal communication with the vent line; and

is disposed proximal to the vent line.

17. The method of claim **15**, further comprising regulating a temperature of the liquid gas fuel using a powered heat source, wherein the powered heat source is in thermal communication with the vent line.

18. The method of claim **15**, further comprising preventing over-pressurizing of the fuel tank using the vent line, wherein the vent line comprises a pressure regulator.

19. The method of claim **15**, wherein the vent line comprises insulation.

20. The method of claim **15**, wherein:
the vent line comprises concentric inner and outer walls
that define a void therebetween; and
the void is maintained under a vacuum.

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